Spectral bounds and their limits in Pauli contextuality: exact configurations and a four-qubit C3 certificate

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A general Lean 4 bound, configurations that attain it, and an exact limit on spectral information in a completion problem

Prior work and scope. The spectral floor and its saturation in the three classical families are due to Jonah [J2] and the works identified in Section 1.2. The C3 completion question, its Jordan representation and Conjecture E are due to Jonah [J3]. This paper combines a formalization and exact configurations attaining the floor with a four-qubit certificate establishing one instance of the failure predicted by Conjecture E. The certificate does not invalidate the spectral floor.

Abstract

Spectral methods give rigorous bounds on Pauli contextuality, but a spectrum need not retain the information required by a related completion problem. We study both aspects in a common setting. For N qubits, $D = 2^N$, and an arbitrary set S of nonidentity Pauli observables, a Lean 4 development proves Jonah's spectral floor: satisfied minus violated commuting triples is at most $(D-2)|S|^{3/2}/(6\sqrt{D-1})$. Saturating configurations outside the classical families have exact contextuality degrees 75 on 135 four-qubit observables, 480 on 496 five-qubit observables, and 729344 on 8128 seven-qubit observables. The first two degrees have kernel-checked counts; the seven-qubit counts are executable exact computations. We relate saturation to pure states with a single nonzero Pauli modulus, two-character sets and flat phase states, including an exhaustive classification of the specified third-level phase-state family through five qubits.

We then exhibit two sign assignments on a 71-point section of the real four-qubit configuration. Their 16×16 Pauli carriers are cospectral, but their 64×64 completion operators have cubic traces 14976 and 15360. This establishes the four-qubit C3 instance of Jonah's Conjecture E: the carrier spectrum alone does not determine the completion spectrum. A block certificate explains the information loss. The largest completion eigenvalues differ by two, and the associated spherical relaxations differ by 32; these are not statements about exact binary completion optima. Lean checks carrier cospectrality and completion non-cospectrality in the actual Pauli model, while the printed factorizations and relaxation gaps remain algebraic results outside that formal scope. Independent exact verifiers, falsification controls and Tsotchke software demonstrations accompany the paper. The work is AI-assisted and is neither externally peer reviewed nor a quantum-hardware experiment.

1. Scope and common question

A noncontextual assignment gives each Pauli observable a fixed value ± 1. Compatible triples impose product constraints, and the contextuality degree counts the fewest constraints any assignment must violate [dB]. A signed sum of Pauli matrices converts this discrete problem into a problem about spectral moments. Its third trace records the satisfied-minus-violated count exactly. Optimizing that moment gives a lower bound on contextuality; special states turn the bound into an exact answer.

A different use of spectral information arises when the assignment is fixed only on a hyperplane section. Completing it on the remaining observables gives a quadratic optimization governed by a second operator. Does the spectrum of the fixed section determine the spectrum of this completion operator? The four-qubit example below shows that it need not: the section spectrum can forget which eigenvalues belong to the two eigenspaces of a distinguished Pauli. The completion depends on that partition. Thus a sharp global bound and a failure of spectral reconstruction coexist without contradiction.

1.1 Contributions and verification levels

The first part formalizes the spectral floor for arbitrary induced Pauli configurations and studies its equality cases. The Lean theorem `hoggar_bound` is stated for the actual complex Pauli matrices, with their multiplication phases and context signs. Finite corollaries include the known exact degrees 63 for three qubits and 315 for the real four-qubit quadric, and stronger lower bounds than $L/5$ for several larger classical configurations. The bound itself is prior work [J2]; the formalization and the extension of its application to arbitrary supports are the contributions considered here.

Section 5 gives exact configurations outside the classical families. The four-qubit pivot support S_B has 135 points, 1095 contexts and degree 75; the five-qubit support S_5 has 496 points, 10880 contexts and degree 480. These degree statements, including their finite counts and witnesses, are proved in Lean. Their interpretation as Pauli supports of particular states is checked separately by exact programs. The seven-qubit circuit support has 8128 points, 2824192 contexts and degree 729344; its finite counts are not kernel-checked. The saturation criterion, quantum-metric interpretation, two-character-set result and classification of specified flat phase states are mathematical and executable results with the boundaries stated in Section 9.

Section 8 supplies a fully specified four-qubit C3 certificate. Two assignments on the 71-point section of the real quadric have equal carrier spectra and different completion spectra. It proves the $N = 4$ C3 instance of the failure predicted by Conjecture E [J3], rather than refuting that conjecture. Lean checks the actual Pauli and signed-context interpretation, not merely an isolated pair of matrices. The polynomial factorizations and the gaps of two and 32 are independently checked algebraic consequences, outside the 42-declaration C3 audit.

The two 135-point configurations must be distinguished. The pivot support S_B is not a quadric and has 1095 contexts and degree 75. The C3 ambient configuration is the real quadric $Q^+(7,2)$, with 1575 contexts and known degree 315 [LGG]; its section leaves 1120 outside contexts. The C3 result concerns completion spectra and does not calculate a new contextuality degree for either set.

Section 9 records the formal boundary for both parts. Sections 10 and 11 give the software demonstrations and falsification checks; Appendices A and B give reproduction commands and witness formats. In the journal layout those four sections form the supplement, retaining their numbers. The full paper and journal forms share the same mathematical statements and named references.

1.2 Relation to prior work

- The contextuality degree and its first values: de Boutray, Holweck, Giorgetti, Masson and Saniga [dB]; Muller, Saniga, Giorgetti, de Boutray and Holweck [M1] (three-qubit degree 63, the split Cayley hexagon of violated contexts); Muller, Saniga, Giorgetti, Holweck and Kelleher [M2] (upper bounds for four to seven qubits, lower bound $L/5$); Saniga et al. [S] (hexagons).
- The exact value 315 for the four-qubit hyperbolic quadric, its 240 optimal sets and a Lean proof of that value: Li, Gupta and Ganesh [LGG].
- The exact values $d(Q^-(7,2)) = 315$, $d(W(7,2)) = 1575$, $d(Q^+(9,2)) = 6975$ (computer-assisted), the operator $A = \sum_v f(v)P_v$ and its Hoggar spectrum $\{21, -3^7\}$: Jonah [J1].
- The spectral floor itself, its Kramers (quaternionic) form for elliptic quadrics, the description of quadrics by the antiunitaries P_cK, the operator identity for A^2 on sets closed under products, and the theorem that among the three families the floor is attained exactly for $W(5,2)$, $Q^+(7,2)$, $Q^-(5,2)$ (and trivially $Q^-(3,2)$): Jonah [J2], with the trace formula also in [J3]. Stronger five-qubit lower bounds ($d(W(9,2)) \geq 28830$, $d(Q^-(9,2)) \geq 6510$) are announced, with proofs held privately, in [J4]. Sets that are not closed under products are named in [J2] as not covered.
- Pure states whose Pauli expectation values are zero or of one modulus: the two-qubit state of Liu, Low and Yin [LLY]; the four-qubit state $(-3, 1, \ldots, 1)/\sqrt{24}$ and the five-qubit state $|s_5\rangle$ with 135 and 496 non-zero values, found numerically by Cieśliński [Ci], who calls a (possibly mixed) state *Pauli flat* when all of its Pauli expectation values have one modulus and lists these pure states under the relaxed condition; real four-qubit states of this kind as the optima of $Q^+(7,2)$ [J4]. Kang [K] proves that no four-qubit state, and no state of an odd number $N > 1$ of qubits, has all squared Pauli expectation values in $\{0, 2^{-N}\}$. None of [LLY, Ci, K] discusses contextuality, phase states, or more than five qubits.
- Diagonal gates of the Clifford hierarchy and their phase polynomials: Cui, Gottesman and Krishna [CGK]. $\mathbb{Z}_4$-valued quadratic forms and their character sums: [CCKS]. Two-character sets, two-weight codes and strongly regular graphs: Calderbank and Kantor [CK]. The bent functions of Maiorana and McFarland: [McF].
- The scalar inequality behind the floor is classical: it is Okumura's lemma [O] (equivalently the maximal sample skewness, Wilkins [W]).
- The Hoggar lines and the cubic purity condition for quantum states: Hoggar [H], Stacey [St], Jones and Linden [JL]. Bent functions and their difference sets: Rothaus [R]. Point sets with the hyperplane intersection numbers of a quadric (quasi-quadrics): De Clerck, Hamilton, O'Keefe and Penttila [DHOP].

The spectral-floor derivation in Sections 3, 4 and 6 was developed before the project identified [J1]–[J4]; the overlap is attributed here. The exhaustive computations behind the exact values of [J1] were not rerun. The assignments found in this project are consistent with those values, which remain external computer-assisted claims.

For completion spectra, [J3] gives the strict-law question, its Jordan representation, the case table and Conjecture E. Its inspected v0.5 reports the three-qubit C3 phenomenon but leaves C3 for $N \geq 4$ open. Section 8 addresses $N = 4$. The C3 literature check examined that primary PDF and related deposits, and searched for the named conjecture and higher-dimensional C3 examples. No additional primary $N \geq 4$ example was identified in that bounded search. This is a qualitative comparison of evidence, not a statistical meta-analysis or a guarantee of historical priority. The source manuscript is dated 20 October 2026, its deposit is dated 21 October, and its latest-version endpoint returned v0.5 when inspected on 9 November.

2. Setting

Pauli operators. Let $D = 2^N$. A *point* is a nonzero vector $v = (a \mid c) \in \mathbb{F}_2^{2N}$. It labels the Hermitian operator

$$P_v \;=\; i^{\,|a\wedge c|}\, X^a Z^c,$$

a tensor product of I, X, Y, Z (the exponent $|a \wedge c|$ counts the factors equal to $Y = iXZ$). For numeric labels use $a + 2^N c$, interpreting each binary vector as an integer with bit zero at the rightmost tensor factor. Words are printed in tensor-factor order from left to right. Thus four-qubit label 128 is $ZIII$. This convention is also used for the C3 witnesses in Section 8. $P_0 = I$, $P_v^2 = I$, and $\operatorname{tr} P_v = 0$ for $v \neq 0$. Products close up to a phase,

$$P_u P_v = \tau(u,v)\, P_{u+v}, \qquad \tau(u,v) = i^{\,k_u+k_v+3k_{u+v}}\,(-1)^{c_u\cdot a_v}, \quad k_v = |a_v \wedge c_v|,$$

and with the symplectic form $\langle u, v\rangle = c_u \cdot a_v + c_v \cdot a_u$ one has $\tau(v,u) = (-1)^{\langle u,v\rangle}\tau(u,v)$ and $\tau(u,v)^2 = (-1)^{\langle u,v\rangle}$. So P_u and P_v commute exactly when $\langle u, v\rangle = 0$, and then $\tau(u,v) = s(u,v) \in \{+1,-1\}$.

Contexts. A *context* (a line of the symplectic polar space $W(2N-1,2)$) is a set $\ell = \{u, v, u+v\}$ of three distinct points with $\langle u, v\rangle = 0$. Its three operators commute and $P_u P_v P_{u+v} = s_\ell\, I$ with $s_\ell = s(u,v)$, which does not depend on the order.

Configurations and assignments. Fix a set S of points and let $\mathcal{L}(S)$ be the set of all contexts contained in S, $L_S = |\mathcal{L}(S)|$. S is *closed* if, for distinct labels $u, v \in S$ with $\langle u, v\rangle = 0$, the label $u + v$ also belongs to S: closure of commuting Pauli products up to sign, excluding the identity product of a label with itself. An assignment $f : S \to \{+1,-1\}$ *violates* ℓ when $f(u)f(v)f(u+v) \neq s_\ell$. Put

$$\beta_S(f) = \sum_{\ell\in\mathcal{L}(S)} s_\ell\, f(u)f(v)f(u+v) = L_S - 2\,\mathrm{viol}(f), \qquad d(S) = \min_f \mathrm{viol}(f).$$

$d(S)$ is the contextuality degree [dB]. For the noncontextuality inequality $\chi = \sum_\ell s_\ell \langle P_u P_v P_{u+v}\rangle \leq L_S - 2d(S)$ quantum mechanics gives $\chi = L_S$ in every state [C, LGG].

Three families. The *full space* is $S = \mathbb{F}_2^{2N}\setminus\{0\}$, with D^2-1 points and $L = (D^2-1)(D^2-4)/12$ contexts. Let $q_0(v) = |a \wedge c| \bmod 2$ for $v = (a \mid c)$. For $w \in \mathbb{F}_2^{2N}$ set $Q_w = \{v \neq 0 : q_0(v) + \langle w, v\rangle = 0\}$. Q_w is a *hyperbolic* quadric $Q^+(2N-1,2)$ if $q_0(w) = 0$ and an *elliptic* quadric $Q^-(2N-1,2)$ if $q_0(w) = 1$. With K complex conjugation in the computational basis, $\theta_w = P_w K$ is antiunitary, $\theta_w P_v \theta_w^{-1} = (-1)^{q_0(v)+\langle w,v\rangle} P_v$ and $\theta_w^2 = (-1)^{q_0(w)}$. So the quadrics are the sets of Pauli observables fixed by a Pauli antiunitary: a real structure ($\theta^2 = 1$) for hyperbolic quadrics and a quaternionic structure ($\theta^2 = -1$) for elliptic ones [J2]. Q_0 is the set of real symmetric Pauli matrices (an even number of Y factors). All three families are closed.

3. The spectral floor

Theorem 1 (spectral floor; Jonah [J2] for the three families; formalized here for arbitrary S). Let $N \geq 1$, let S be any set of points and $f : S \to \{\pm 1\}$. Then

$$\beta_S(f) \;\leq\; \frac{(D-2)\,|S|^{3/2}}{6\sqrt{D-1}}, \qquad \text{hence} \qquad d(S) \;\geq\; \frac{L_S}{2} - \frac{(D-2)\,|S|^{3/2}}{12\sqrt{D-1}}.$$

Proof. Let $A = \sum_{v\in S} f(v) P_v$, a Hermitian $D \times D$ matrix.

(i) Three traces. $\operatorname{tr} A = 0$ because $0 \notin S$. Since $\operatorname{tr}(P_u P_v) = D\,\delta_{uv}$ and $f(v)^2 = 1$, $\operatorname{tr} A^2 = D\,|S|$. Since $\operatorname{tr}(P_u P_v P_w) = D\,\tau(u,v)\,[w = u+v]$,

$$\operatorname{tr} A^3 = D \sum_{\substack{u,v \in S \\ u+v \in S}} f(u) f(v) f(u+v)\,\tau(u,v).$$

The pairs with $\langle u, v\rangle = 1$ cancel against their transposes, because $\tau(v,u) = -\tau(u,v)$. The pairs with $\langle u, v\rangle = 0$ are the ordered pairs of distinct points of a context ($u = v$ would give $u + v = 0 \notin S$), six for each context, each contributing $s_\ell f(u) f(v) f(u+v)$. Therefore $\operatorname{tr} A^3 = 6D\,\beta_S(f)$.

(ii) A scalar inequality (Okumura [O]). Let $x_1, \dots, x_n$ be real with $\sum x_i = 0$ and $\sum x_i^2 = nm$, $n \geq 2$, and let $a = \sqrt{(n-1)m}$. By the Cauchy–Schwarz inequality applied to the other $n-1$ numbers, $x_i^2 \leq (n-1)(nm - x_i^2)$, so $x_i \leq a$ for every i. Hence $(x_i - a)\big((n-1)x_i + a\big)^2 \leq 0$. Summing over i and using the two constraints,

$$(n-1) \sum_i x_i^3 \;\leq\; n\,m\,(n-2)\,a.$$

(iii) Conclusion. Apply (ii) to the eigenvalues of A with $n = D$, $m = |S|$: $(D-1) \cdot 6D\beta \leq D\,|S|\,(D-2)\sqrt{(D-1)|S|}$. ∎

Three remarks. First, no closure property of S is used: products that leave S do not contribute to the trace. Second, the hypothesis $f = \pm 1$ enters only through $\operatorname{tr} A^2 = D|S|$, so the argument bounds the cubic form $T(x) = \sum_\ell s_\ell x_u x_v x_w$ on the sphere $\sum x_v^2 = |S|$. Third, the bound applies to the induced configuration only, that is, with *all* contexts contained in S.

Theorem 2 (quaternionic form; Jonah [J2]). Let $N \geq 2$. If every P_v, $v \in S$, commutes with an antiunitary θ with $\theta^2 = -1$ (for instance, S lies in an elliptic quadric), then

$$\beta_S(f) \;\leq\; \frac{(D-4)\,|S|^{3/2}}{6\sqrt{2(D-2)}}.$$

Proof. A is a real combination of operators commuting with θ, so $\theta A = A\theta$. If $Ax = \lambda x$ then $A\theta x = \lambda \theta x$, and $\langle x, \theta x\rangle = 0$ because $\langle \theta x, \theta y\rangle = \overline{\langle x, y\rangle}$ gives $\langle x, \theta x\rangle = \overline{\langle \theta x, \theta^2 x\rangle} = -\langle x, \theta x\rangle$. Every eigenspace is therefore an orthogonal sum of planes $\{x, \theta x\}$ and every eigenvalue has even multiplicity (Kramers). Apply (ii) to one value from each of the $D/2$ pairs, counted with pair multiplicity, with the same m. ∎

Theorem 2 is not part of the Lean development; it is checked numerically (Section 11).

Closed forms. With the standard counts, Theorems 1 and 2 give for the three families the same expression in different arguments. Writing $\varphi(D) = \frac{1}{2} - \sqrt{D+1}/(D+2)$,

the fraction column below is understood only when $L > 0$; configurations without contexts have degree zero, and no violated fraction is defined for them.

family	points	contexts L	violated fraction at least
full space $W(2N-1,2)$	D^2-1	$(D^2-1)(D^2-4)/12$	$\varphi(D)$
elliptic quadric $Q^-(2N-1,2)$	$(D+1)(D-2)/2$	$(D+1)(D-2)(D+2)(D-4)/48$	$\varphi(D)$
hyperbolic quadric $Q^+(2N-1,2)$	$(D-1)(D+2)/2$	$(D-1)(D+2)(D-2)(D+4)/48$	$\varphi(D/2)$

The three cases are the complex, quaternionic and real members of Dyson's threefold way; the real family at N qubits has the same floor as the complex family at $N-1$ qubits.

4. Saturation in the classical families

Theorem 3 (equality; Jonah [J2, Theorem 6] for the three families). Suppose $S \neq \varnothing$. Equality holds in Theorem 1 if and only if there is a unit vector ψ with

$$\langle\psi|P_v|\psi\rangle = f(v)\sqrt{\frac{D-1}{|S|}} \quad (v \in S), \qquad \langle\psi|P_v|\psi\rangle = 0 \quad (v \notin S,\ v \neq 0).$$

Proof. Equality in (ii) forces $x_i \in \{a, -a/(n-1)\}$ for every i, and the zero sum forces exactly one value a. So $A = (a-b)\,|\psi\rangle\langle\psi| + b\,I$ with $b = -a/(D-1)$; reading off Pauli coefficients gives the claim, and the converse is the same computation backwards. ∎

We call such a ψ *flat on* S, after [Ci]. For the full space the condition says that ψ is a fiducial vector of a SIC covariant under the N-qubit Pauli group. Such vectors exist only for $N = 1$ and $N = 3$ (Godsil and Roy [GR]; an elementary proof through the integrality of the floor is in [J2]): for $N = 3$ they are the Hoggar lines.

The three saturated cases. The Hoggar fiducial $\psi_H \propto (-1+2i, 1, 1, 1, 1, 1, 1, 1)$ has $\langle P_v\rangle = \pm 1/3$ for all 63 three-qubit Pauli observables. We verified the following by exact counting in four independent implementations (Section 11); the values themselves are known [M1, dB, LGG, J2].

configuration	assignment	violated	floor	spectrum of A
$W(5,2)$: 63 points, 315 contexts	signs of $\langle\psi_H\|P_v\|\psi_H\rangle$	63	63	$\{21, (-3)^7\}$
$Q^-(5,2)$: 27 points, 45 contexts	the same signs, restricted	9	9	$\{9^2, (-3)^6\}$
$Q^+(7,2)$: 135 points, 1575 contexts	signs of the realified $\psi_H \in \mathbb{R}^{16}$	315	315	$\{45, (-3)^{15}\}$

In the third row, $\mathbb{C}^8$ is read as $\mathbb{R}^{16}$; the resulting real four-qubit state has expectation $\pm 1/3$ on all 135 real Pauli observables and 0 on the 120 imaginary ones (this description of the optimal assignments of $Q^+(7,2)$ as sign patterns of real flat states is announced in [J4]; the state $(-3, 1, \ldots, 1)/\sqrt{24}$ of [Ci] is of this kind). The violated contexts form, respectively, a classically embedded split Cayley hexagon (63 lines, three per point, incidence graph of diameter 6 and girth 12, the three lines through each point in one totally isotropic plane), nine pairwise disjoint lines, and a copy of the dual polar space $DW(5,2)$ (315 lines, seven per point, collinearity graph distance-regular with intersection array $\{14, 12, 8; 1, 3, 7\}$), in agreement with [M1, S, M2, LGG].

By Theorem 3, the optimal assignments of $W(5,2)$ are exactly the sign patterns of Pauli-covariant SIC fiducials in dimension 8. Our enumeration finds 120 optimal violated sets in the orbit of the Hoggar pattern under $Sp(6,2)$, and the context–point incidence matrix has rank 56 over $\mathbb{F}_2$, so each set comes from $2^7 = 128$ assignments: $120 \cdot 128 = 15360 = 240 \cdot 64$ assignments, matching the 240 Hoggar-type SICs of 64 vectors each [St]. For $Q^+(7,2)$ the orbit of our violated set under the reflections of $O^+(8,2)$ has 240 elements, the number proved in [LGG].

Proposition 3 (violated contexts in the quantum metric). Let ψ be a state, $p_v = \langle P_v\rangle$, and let $Q_{uv} = \frac{1}{4}\big(\langle P_u P_v\rangle - p_u p_v\big)$ be its quantum geometric tensor along the flows $e^{-itP_v/2}$. Consider a context $\{u, v, w\}$ with sign s and assume $p_u p_v p_w \neq 0$, so the sign assignment on this context is defined. The Fubini-Study metric is $g_{uv} = \frac{1}{4}(s\,p_w - p_u p_v)$. If $f = \mathrm{sign}(p)$, the context is violated exactly when $4|g_{uv}| = |p_w| + |p_u p_v|$ and satisfied exactly when $4|g_{uv}| = \big||p_w| - |p_u p_v|\big|$.

Proof. $\langle P_u P_v \rangle = s\, p_w$; the context is violated when $s\, p_w$ and $p_u p_v$ have opposite signs. ∎

For ψ_H: $g_{vv} = 2/9$; $|g_{uv}| = 1/9$ on the 189 commuting pairs lying on the 63 violated contexts, $1/18$ on the 756 pairs lying on satisfied contexts, $1/36$ on the 1008 anticommuting pairs; the Berry curvature has modulus $1/6$ on anticommuting pairs and vanishes on commuting ones. The hexagon is the set of pairs of largest off-diagonal metric (computed with the QGTL kernels, Section 10).

5. Saturation outside the classical families

5.1 The criterion

The statements of this section concern arbitrary sets S, which need not be closed. For an assignment f and a point $w \in S$ let the *excess* $e_w(f)$ be the number of satisfied minus the number of violated contexts of S through w. For a point $w \notin S$, $w \neq 0$, let the *outside sum* be $b_w(f) = \sum s(u,v) f(u) f(v)$ over the unordered pairs of commuting points u, v of S with $u + v = w$.

Theorem 4 (saturation criterion). Let S be a nonempty set of points, $f : S \to \{\pm 1\}$ and $A = \sum_{v \in S} f(v) P_v$. Then

$$A^2 \;=\; |S|\, I \;+\; 2 \sum_{w \in S} e_w(f)\, f(w)\, P_w \;+\; 2 \sum_{w \notin S,\; w \neq 0} b_w(f)\, P_w,$$

and the following are equivalent:

- (a) f attains the floor: $\beta_S(f) = (D-2)|S|^{3/2}/(6\sqrt{D-1})$;
- (b) A has the simple eigenvalue $\sqrt{(D-1)|S|}$ and the eigenvalue $-\sqrt{|S|/(D-1)}$ with multiplicity $D-1$;
- (c) some pure state is flat on S with signs f (Theorem 3);
- (d) $e_w(f) = (D-2)\sqrt{|S|}\,/\,(2\sqrt{D-1})$ for every $w \in S$, and $b_w(f) = 0$ for every $w \notin S$ with $w \neq 0$.

When they hold, $d(S) = \frac{1}{2} L_S - (D-2)|S|^{3/2}/(12\sqrt{D-1})$ exactly, f is optimal, and, for $N \geq 2$, $(D-1)|S|$ is a perfect square.

Proof. In $A^2 = \sum_{u,v} f(u) f(v) P_u P_v$ the diagonal terms give $|S|\, I$ and the two orders of an anticommuting pair cancel. A commuting pair $u \neq v$ contributes $2 s(u,v) f(u) f(v) P_{u+v}$. If $w = u + v \in S$, the pair is a context through w and $s(u,v) f(u) f(v) = \pm f(w)$ according to whether it is satisfied; summing over the contexts through w gives $2 e_w(f) f(w) P_w$. If $w \notin S$ the pairs give $2 b_w(f) P_w$. This is the identity; for closed S the last sum is empty and it is Theorem 1 of [J2]. (a) ⇔ (b) is the equality case of the scalar inequality, and (b) ⇔ (c) is Theorem 3. Condition (b) says $(A - a)(A + a/(D-1)) = 0$ with $a = \sqrt{(D-1)|S|}$, that is $A^2 = |S|\, I + \frac{a(D-2)}{D-1} A$ (a traceless matrix with these two eigenvalues has the larger one exactly once); comparing Pauli coefficients with the identity gives (d). The floor is an integer when attained, which for $D > 2$ forces $\sqrt{(D-1)|S|}$ to be rational. ∎

Condition (d) replaces the *violation-regularity* of [J2, Theorem 6], to which it reduces when every point lies on the same number of contexts. For $N \leq 5$ the number $D - 1$ is squarefree, so a set that attains the floor has $|S| = (D-1)k^2$ and excess $(D-2)k/2$: $|S| \in \{3, 12\}$ for two qubits, $\{7, 28, 63\}$ for three, $\{15, 60, 135, 240\}$ for four, $\{31, 124, 279, 496, 775\}$ for five.

Table 1. The configurations we know to attain the floor (besides the stabilizer case $|S| = D - 1$, $d = 0$, a maximal commuting set with the signs of a stabilizer state).

state	N	observables	contexts	degree	modulus	excess	closed	state from
$(i,i,i,1)/2$	2	12	8	0	1/2	2	no	[LLY]
CCZ applied to $(1,\dots,1)/\sqrt{8}$	3	28	56	0	1/2	6	no	folklore
Hoggar fiducial	3	63	315	63	1/3	9	yes	[H]
$(-3,1,\dots,1)/\sqrt{24}$	4	135	1575	315	1/3	21	yes	[J4, Ci]
ψ_B (pivot state)	4	135	1095	**75**	1/3	21	no	this work
s_5	5	496	10880	**480**	1/4	60	no	[Ci]
Φ_5 (3 CS, 2 CCZ)	5	496	10880	**480**	1/4	60	no	this work
Φ_7 (3 CS, 8 CCZ)	7	8128	2824192	**729344**	1/8	504	no	this work

The degrees in bold are new, as is the reading of the first two rows: the 12 two-qubit observables outside a maximal commuting set, and the 28 real three-qubit observables that are not diagonal, are noncontextual configurations on which the floor is nevertheless exact. The configurations of s_5 and Φ_5 are equivalent under the Clifford group (Theorem 12). The exact-degree statements for the first seven masks, with their finite counts, are proved in Lean (Section 9); their state identifications are checked separately. All eight states are reproduced on Moonlab (Section 10).

5.2 A four-qubit and a five-qubit configuration

Theorem 5 (the pivot configuration). Let

$$\psi_B \;=\; \tfrac{1}{\sqrt{6}}\big(\,|0011\rangle + |0111\rangle + |1011\rangle + |1100\rangle + |1101\rangle + i\,|1110\rangle\,\big),$$

with qubit 0 written on the right, and let S_B be the set of observables with non-zero expectation value in ψ_B.

1. The expectation values are $\pm 1/3$ on S_B, and $|S_B| = 135$: the 15 diagonal observables $(0 \mid c)$, and for each $a \neq 0$ the eight observables $(a \mid c)$ with $a \cdot c = \epsilon(a)$, where $\epsilon(a) = 1$ for the five values $a \in E = \{0010, 0011, 0101, 1001, 1101\}$ and $\epsilon(a) = 0$ otherwise.
2. S_B contains 1095 contexts: 35 through each diagonal observable and 23 through each of the other 120. It is not closed and it is not a quadric, but every hyperplane $w^\perp$ meets it in 71 points if $w \in S_B$ and in 63 points otherwise, as for $Q^+(7,2)$: it is a quasi-quadric in the sense of [DHOP]. It differs from the quadric $a \cdot c = 0$ in 80 points, those with $a \in E$, and E consists of five pairwise non-orthogonal vectors (an ovoid) of the doily $W(3,2)$ defined by the form $x_0y_1 + x_1y_0 + x_2y_3 + x_3y_2$.
3. $d(S_B) = 75$. The assignment given by the signs of the expectation values violates the 15 lines of that doily, read among the diagonal observables, and 60 further contexts, each with exactly one diagonal observable. Every non-diagonal observable of S_B lies on exactly one violated context and every diagonal one on seven.

Proof. Items 1 and 2 and the count of 75 violated contexts are finite computations in $\mathbb{Z}[i]$ (Section 11); item 1 also follows from Proposition 6. The bound $d(S_B) \geq 75$ is Theorem 1 with $|S| = 135$, $D = 16$, $L = 1095$: $\beta \leq 14 \cdot 135 \cdot 45/(6 \cdot 15) = 945$. In Lean this is `pivot_degree`. ■

The noncontextuality inequality of this configuration has classical bound 945 and quantum value 1095 in every state. The quadric $Q^+(7,2)$ has the same number of observables and the same classical bound, with quantum value 1575.

Proposition 6 (flat states on a bent support). Let T be the set of the six $x \in \mathbb{F}_2^4$ with $x_0x_1 + x_2x_3 = 1$, the support of a bent function, and let $\phi : T \to \mathbb{C}$ with $|\phi| = 1$. The state

$\psi_\phi = 6^{-1/2}\sum_{x\in T}\phi(x)|x\rangle$ is flat on its Pauli support if and only if, up to a global phase, all $\phi(x)$ are fourth roots of unity. In that case the support has 135 points and the modulus is $1/3$; it is a hyperbolic quadric (1575 contexts, degree 315) when the number of imaginary amplitudes is even, and it has 1095 contexts and degree 75 when that number is odd. The two classes are not related by any Clifford operation.

Proof. T is a difference set with parameters $(16, 6, 2)$ [R]: every non-trivial character sum over T is ± 2, and for every $a \neq 0$ exactly one unordered pair $x_a, x_a + a$ lies in T. Hence $\langle Z^c\rangle = \pm 1/3$ for $c \neq 0$, and for $a \neq 0$, with $z_a = \overline{\phi(x_a + a)}\,\phi(x_a)$,

$$\langle X^a Z^c\rangle = \tfrac{1}{6}(-1)^{c\cdot x_a}\big(z_a + (-1)^{c\cdot a}\,\overline{z_a}\big),$$

so the expectation value of $P_{(a|c)}$ is $\pm\frac{1}{3}\operatorname{Re} z_a$ when $a\cdot c = 0$ and $\pm\frac{1}{3}\operatorname{Im} z_a$ when $a\cdot c = 1$. A common modulus must be $1/3$, which holds exactly when every z_a is ± 1 or $\pm i$, that is, when all ratios of amplitudes are fourth roots of unity. Then for each $a \neq 0$ exactly the eight observables with $a\cdot c = \epsilon(a)$ are in the support, where $\epsilon(a) = 0$ if z_a is real and 1 if it is imaginary. Write $\phi = i^k$ and $\kappa = k \bmod 2$; then $\epsilon(x + x') = \kappa_x + \kappa_{x'}$. A diagonal Clifford operation multiplies ϕ by $i^{q(x)}$ for a $\mathbb{Z}_4$-valued quadratic form q and changes κ by the restriction to T of a linear functional; these sixteen restrictions have weights 0, 2 and 4, and together with the all-ones vector (a global factor i) they span the even-weight vectors of $\mathbb{F}_2^T$. So κ of even weight is equivalent to $\kappa = 0$, a real state, whose support is the quadric $a\cdot c = 0$; and κ of odd weight is equivalent to a single imaginary amplitude, as in ψ_B. Symplectic maps preserve the number of contexts, and $1575 \neq 1095$. The two counts were computed for all 32 supports that occur, over all 4096 phase patterns (Section 11). ∎

Theorem 7 (five qubits). Let S_5 be the set of the 496 five-qubit observables with non-zero expectation value in the state s_5 of Cieśliński [Ci, eq. (18)], all equal to $\pm 1/4$. Then S_5 contains 10880 contexts and $d(S_5) = 480$. The optimal assignment, the signs of the expectation values, leaves 16 observables on no violated context (60 contexts through each), 240 on two (of 64) and 240 on four (of 68). Every hyperplane $w^\perp$ meets S_5 in 256 points if $w \in S_5$ and in 240 otherwise, as for the complement of $Q^+(9, 2)$; S_5 is not such a complement.

Proof. The counts are finite computations in $\mathbb{Z}[i]$. The lower bound is Theorem 1 with $|S| = 496$, $D = 32$: $\beta \leq 30{\cdot}496{\cdot}124/(6{\cdot}31) = 9920 = 10880 - 2{\cdot}480$. In Lean this is `cieslinski5_degree`. No state vanishes on every observable of a hyperbolic quadric: these are the observables commuting with an antiunitary θ with $\theta^2 = 1$, and the condition would give $\rho + \theta\rho\,\theta^{-1} = 2I/D$, of rank D, for a projector ρ of rank one. ∎

Proposition 8 (the optimum in the quantum metric). Let ψ be flat on S with modulus μ, and let f be the signs of its expectation values. For $u \neq v$ in S the Fubini–Study metric of Proposition 3 has $|g_{uv}| = (\mu + \mu^2)/4$ if $u, v, u + v$ form a context of S violated by f, $|g_{uv}| = (\mu - \mu^2)/4$ if they form a satisfied context of S, and $|g_{uv}| = \mu^2/4$ otherwise (u, v anticommute, or their product is outside S). So the violated contexts of the optimal assignment are exactly the pairs with $4|g_{uv}|$ larger than μ. The Berry curvature has modulus $\mu/2$ on anticommuting pairs whose product is in S and vanishes on all other pairs.

Proof. Proposition 3 with $|p| = \mu$ on S and $p = 0$ outside; for anticommuting u, v the product $P_u P_v$ is $\pm i P_{u+v}$, so $\langle P_u P_v\rangle$ is imaginary, of modulus $|p_{u+v}|$. ∎

Proposition 9 (two mutually unbiased stabilizer states). Let s_1, s_2 be stabilizer state vectors whose underlying Pauli label spaces are complementary: they intersect only in the zero label. Then $|\langle s_1|s_2\rangle| = D^{-1/2}$, and let $\varphi = \theta + \arg\langle s_1|s_2\rangle$. The normalized state proportional to $s_1 + e^{i\theta}s_2$ is flat on a set that is not a stabilizer group exactly in the following cases: $N = 1$ with $\cos\varphi = (\sqrt{6} - \sqrt{2})/4$ or $-(\sqrt{6} + \sqrt{2})/4$ (all 3 observables, modulus $1/\sqrt{3}$); $N = 2$ with $\varphi = \pm\pi/2$ (12 observables, modulus $1/2$); $N = 3$ with $\varphi = \pm 3\pi/4$ (all 63, modulus $1/3$: the

Hoggar fiducial is $\sqrt{8}$ times the uniform vector plus $(-2+2i)$ times the first basis vector); $N = 4$ with $\varphi = \pi$ (the 135 observables of a hyperbolic quadric, modulus 1/3: $(-3, 1, \ldots, 1)$ is 4 times the uniform unit vector minus 4 times the first basis vector). There is none for $N \geq 5$.

Proof. Let $t = D^{-1/2}$ and $n = 2+2t\cos\varphi$ the squared norm. An observable of the first stabilizer group has expectation $\pm(1 + 2t\cos\varphi)/n$, and so has one of the second. Every other observable is, up to a phase, a product P_1P_2 of non-identity elements of the two groups, and its expectation is $\pm 2t\cos\varphi/n$ if P_1, P_2 commute and $\pm 2t\sin\varphi/n$ if they anticommute. Equating the non-zero moduli: all three are equal only if $|\cos\varphi| = |\sin\varphi|$ and $1-\sqrt{2}\,t = \sqrt{2}\,t$, so $D = 8$; with $\sin\varphi = 0$ one needs $1 - 2t = 2t$, so $D = 16$ and $\cos\varphi = -1$; with $\cos\varphi = 0$ one needs $1 = 2t$, so $D = 4$; for $D = 2$ there are no commuting products and $|1 + \sqrt{2}\cos\varphi| = \sqrt{2}|\sin\varphi|$ has the two stated solutions. The remaining case $1 + 2t\cos\varphi = 0$ gives a stabilizer state. ■

So the SIC fiducials of one and three qubits, a state with the Pauli spectrum of [LLY], and the quadric state are four members of one family, and that family stops at four qubits. The pivot state is not in it: it is a combination of three pairwise unbiased stabilizer states.

5.3 Flat phase states, and a seven-qubit configuration

Proposition 10 (the support is a two-character set). Let ψ be flat on S and let $k^2 = |S|/(D-1)$, the inverse square of the modulus. For every point w,

$$\sum_{v\in S}(-1)^{\langle v,w\rangle} \;=\; D\,[w \in S] \;-\; k^2.$$

Equivalently, exactly $(D-2)|S|/(2(D-1)) + \frac{D}{2}[w \in S]$ observables of S commute with P_w. Hence the graph on $\mathbb{F}_2^{2N}$ in which u and v are adjacent when $u + v \in S$ is strongly regular (possibly imprimitive, or complete), with degree $|S|$ and nonprincipal character eigenvalues among $D - k^2$ and $-k^2$ [CK]; an empty character class contributes no eigenvalue. If $k^2 = D/2$ the sum takes the values $\pm D/2$: then S is a Hadamard difference set, the support of a bent function on the $2N$-dimensional space.

Proof. For a pure state ρ and any w, $\mathrm{tr}(\rho P_w \rho P_w) = \langle P_w\rangle^2$. Writing $\rho = D^{-1}\sum_v p_v P_v$ with $p_0 = 1$ and using $P_wP_vP_w = (-1)^{\langle v,w\rangle}P_v$ gives $p_w^2 = D^{-1}\sum_v(-1)^{\langle v,w\rangle}p_v^2$. Insert $p_v^2 = k^{-2}$ on S, 1 at $v = 0$ and 0 elsewhere. ■

The identity for p_w^2 holds for every pure state and is standard; we did not find its consequence for flat states stated. It accounts for the hyperplane numbers in Theorems 5 and 7. For the two four-qubit rows of Table 1 it gives a strongly regular graph with the parameters $(256, 135, 70, 72)$ of the affine polar graph of $Q^+(7, 2)$; for the three-, five- and seven-qubit rows with modulus $2^{-(N-1)/2}$ it gives bent functions in 6, 10 and 14 variables.

Phase states. Let $a_j \in \mathbb{Z}_8$, $b_{jk} \in \mathbb{Z}_4$, $c_{jkl} \in \mathbb{Z}_2$ and

$$p(x) = \sum_j a_j x_j + 2\sum_{j<k} b_{jk}\,x_jx_k + 4\sum_{j<k<l} c_{jkl}\,x_jx_kx_l \pmod 8, \qquad |\Phi_p\rangle = D^{-1/2}\sum_x e^{i\pi p(x)/4}\,|x\rangle.$$

Φ_p is obtained from $|+\cdots+\rangle$ by T gates (a_j), controlled-S gates CS (b_{jk}) and CCZ gates (c_{jkl}); these are the diagonal gates of the third level of the Clifford hierarchy [CGK]. Let T be the symmetric trilinear form on $\mathbb{F}_2^N$ with $T(e_j, e_j, e_j) = a_j$, $T(e_j, e_j, e_k) = T(e_j, e_k, e_k) = b_{jk}$ and $T(e_j, e_k, e_l) = c_{jkl}$ modulo 2. Then $B(x, z) = T(x, x, z)$ is a symmetric bilinear form, alternating exactly when there is no T gate. A linear change of variables in T is a CNOT circuit on the state, and two polynomials with the same form differ by a diagonal Clifford gate.

Proposition 11 (phase states of the third level). For $a \neq 0$ let R_a be the radical and r_a the rank of the symmetric bilinear form $T(a,\cdot,\cdot)$. The expectation value of X^aZ^c in Φ_p has modulus $2^{-r_a/2}$ for the 2^{r_a} vectors c of one coset of $R_a^{\perp}$ and vanishes for every other c; diagonal observables have expectation value zero. So Φ_p has a single non-zero modulus exactly when $r_a = r$ is the same for every $a \neq 0$. Then the modulus is $2^{-r/2}$, the support has $(D-1)2^r$ points, r is even if $N \geq 2$, and Φ_p attains the floor on its support. If moreover $r = N-1$, each R_a has one non-zero element $\rho(a)$, the map ρ is an involution of the non-zero vectors, and

$$S = \{\, (a \mid c) \;:\; a \neq 0, \quad c \cdot \rho(a) = \epsilon(a) \,\}$$

for a function ϵ: the support of a bent function of Maiorana–McFarland type [McF].

Proof. Modulo 8 the third difference of p is $4T$, independent of the base point, and the second difference is even. So $p(x) - p(x+a) = h_a + 2q_a(x)$ with $q_a(0) = 0$ and $q_a(x+y) = q_a(x) + q_a(y) + 2T(a,x,y)$ modulo 4, a $\mathbb{Z}_4$-valued quadratic form [CCKS], and $\langle X^aZ^c\rangle = D^{-1}e^{i\pi h_a/4}\sum_x i^{q_a(x)}(-1)^{c\cdot x}$. Now

$$\Big|\sum_x i^{q_a(x)}(-1)^{c\cdot x}\Big|^2 = \sum_z i^{q_a(z)}(-1)^{c\cdot z}\sum_y (-1)^{T(a,y,z)} = D\sum_{z\in R_a} i^{q_a(z)}(-1)^{c\cdot z}.$$

On R_a the map $z \mapsto i^{q_a(z)}$ is a character with values ± 1, so the last sum is 2^{N-r_a} when $(-1)^{c\cdot z}$ agrees with it on R_a and 0 otherwise. That r is even follows from Theorem 4. If $r = N-1$ then $T(a,\rho(a),\cdot) = 0$, which is symmetric in a and $\rho(a)$, so $\rho(\rho(a)) = a$. ∎

Theorem 12 (all flat phase states of the third level on at most five qubits). Among the non-zero forms T in $N \leq 5$ variables, those of constant rank are the following.

N	forms	of constant rank	classes under CNOT circuits	gates of a representative	r	observables
1	1	1	1	T	1	2
2	7	1	1	$CS(0,1)$	2	12
3	127	8	2	$CCZ(0,1,2)$; $CS(0,1)\ CS(0,2)$	2	28
4	16383	0				
5	33554431	166656	1	$CS(0,1)\ CS(1,3)\ CS(2,4)$ $CCZ(0,1,4)\ CCZ(0,2,3)$	4	496

For five qubits the stabilizer of the form in $GL(5,2)$ has order 60. Write Φ_5 for the state of the last row. The supports of the two three-qubit classes are equivalent under $Sp(6,2)$, and the support of Φ_5 is mapped onto the set S_5 of Theorem 7 by a symplectic map of $\mathbb{F}_2^{10}$. Thus no state with nonzero associated form T in this family is flat on four qubits; on five qubits every flat state with nonzero T is obtained from Φ_5 by a CNOT circuit and a diagonal Clifford gate; and the configuration found numerically in [Ci] is, up to a Clifford operation, the Pauli support of a circuit of five gates. The excluded zero form gives diagonal-Clifford stabilizer states, which are flat in every number of qubits.

Proof. Exhaustive enumeration, with the orbits computed from the transvections that generate $GL(N,2)$ (`src/third_level.c`; for $N \leq 4$ also `scripts/phase_states.py`). The two symplectic maps are found by search and verified on every point; the second is recorded in `results/phase_states.json`. ∎

For $N = 2$ the state is that of [LLY] up to a phase and complex conjugation. In coordinates $X = (x_1, x_2, \alpha) \in \mathbb{F}_q \times \mathbb{F}_q \times \mathbb{F}_2$ with $q = 4$, the five-qubit form is

$$T(X,Y,Z) = \mathrm{Tr}\left[x_1y_1z_2^2 + x_1y_2^2z_1 + x_2^2y_1z_1 + x_2y_2z_1^2 + x_2y_1^2z_2 + x_1^2y_2z_2\right] + \alpha\,\sigma(y,z) + \beta\,\sigma(x,z) + \gamma\,\sigma(x,y),$$

where $\sigma(y,z) = \mathrm{Tr}[a\,(y_1z_2 + y_2z_1)]$, Tr is the trace to $\mathbb{F}_2$ and a generates $\mathbb{F}_4$. It is invariant under $(x_1, x_2) \mapsto (\lambda x_1, \lambda^{-1}x_2)$.

Theorem 13 (seven qubits). Let

$$|\Phi_7\rangle = CS_{01}\,CS_{23}\,CS_{45}\;\;CCZ_{012}\,CCZ_{035}\,CCZ_{124}\;\;CCZ_{046}\,CCZ_{056}\,CCZ_{136}\,CCZ_{236}\,CCZ_{256}\;|+\rangle^{\otimes 7}$$

and let S_7 be its Pauli support.

1. Φ_7 has expectation values $\pm 1/8$ on S_7 and 0 elsewhere, and $|S_7| = 8128 = 2^6(2^7 - 1)$.
2. S_7 contains 2 824 192 contexts and $d(S_7) = 729\,344$. In the optimal assignment every observable of S_7 lies on 504 more satisfied than violated contexts; the number of contexts through an observable is 1008, 1024, 1040, 1056, 1072 or 1104 (for 64, 3584, 1792, 1344, 896 and 448 observables).
3. Let $\mathcal{F}$ be the family of forms on $\mathbb{F}_2^7$ for which B is alternating of rank 6. The forms of constant rank in $\mathcal{F}$ are a single class under CNOT circuits, and the stabilizer of such a form in $GL(7,2)$ has order 7. In coordinates $\mathbb{F}_8 \times \mathbb{F}_8 \times \mathbb{F}_2$ the form is the one displayed above, with the trace of $\mathbb{F}_8$, with $a^3 + a^2 + 1 = 0$, and with the additional term $\mathrm{Det}(x_1, y_1, z_1)$, the alternating trilinear form of $\mathbb{F}_8$ over $\mathbb{F}_2$.
4. For each of the ten other types of B (alternating of rank 0, 2 or 4; not alternating, of rank 1 to 7) we tested 3×10^8 random forms; none has constant rank.

Proof. Item 1 and the counts of item 2 are finite computations in $\mathbb{Z}[i]$, made by three programs that share no code (Section 11). The lower bound is Theorem 1 with $|S| = 8128$, $D = 128$: $\beta \le 126 \cdot 8128 \cdot 1016/(6 \cdot 127) = 1\,365\,504 = 2\,824\,192 - 2 \cdot 729\,344$. For item 3, choose a basis in which e_6 spans the radical of B and $e_0, \ldots, e_5$ is a symplectic basis of a complement. Every form of $\mathcal{F}$ is then the third difference of $2(x_0x_1 + x_2x_3 + x_4x_5) + 4w(x_0, \ldots, x_5) + 4x_6\,s(x_0, \ldots, x_5)$ for a set w of CCZ gates on the first six qubits (2^{20} choices) and a graph s on them, which must be nondegenerate (13888 choices). All 1.46×10^{10} such forms were tested (`src/family7.c`): 13 271 040 have constant rank. The group that preserves the normal form has order $64\,|Sp(6,2)| = 92\,897\,280$, and the solutions are one orbit of it, of size $92\,897\,280/7$ (`src/orbits7.c`). ∎

The noncontextuality inequality of S_7 has classical bound 1 365 504 and quantum value 2 824 192: no noncontextual model reproduces more than 74.2 per cent of its contexts. Item 4 is evidence, not proof, that Φ_7 represents the only class of flat phase states with nonzero associated form on seven qubits, up to CNOT circuits and diagonal Clifford gates. The numbers of observables for three, five and seven qubits, 28, 496 and 8128, are $2^{N-1}(2^N - 1)$; that they are the second, third and fourth perfect numbers reflects only that 7, 31 and 127 are prime. We do not know whether this family extends to a nine-qubit flat phase state of constant rank eight, which would have 130816 observables of modulus 1/16. Proposition 11 does not rule out other constant even ranks. The tested forms with the symmetry $(x_1, x_2) \mapsto (\lambda x_1, \lambda^{-1}x_2)$ of the five- and seven-qubit ones do not have constant rank for nine or eleven qubits.

5.4 What does not attain the floor

For the $D^2 - D$ observables outside a maximal commuting set (the non-diagonal ones, say) there are $(D-1)(D-2)D^2/12$ contexts and Theorem 1 gives a violated fraction of at least $\frac{1}{2} - 2^{-N/2}$.

By Theorem 4, equality would need a state with all non-diagonal expectation values of modulus $2^{-N/2}$, that is, a state whose orbit under the Pauli group forms, with the computational basis, a complete set of mutually unbiased bases. For two qubits this is the first row of Table 1. For four qubits and for every odd $N \geq 3$ no such state exists [K], so the degree is strictly above the floor there. A numerical search from random starts found flat four-qubit states only with $|S| = 135$ (both classes of Proposition 6) and flat five-qubit states only with $|S| = 496$; it found none with $|S| = 60$, or with $|S|$ equal to 124, 279 or 775. This is evidence, not proof, that Table 1 is complete up to five qubits.

6. Consequences for all N

Corollary A (full space). For every $N \geq 1$ every assignment of the $D^2 - 1$ Pauli observables violates at least $\varphi(D)\, L$ contexts (Lean: `full_space_fraction`). Since a uniformly random assignment violates $L/2$ contexts on average, $d/L \to 1/2$.

The count inequality also covers $N = 1$, when there are no contexts. The ratios and noise thresholds below are stated for $N \geq 2$, so $L > 0$.

Exponential violation. The ratio of the quantum value to the noncontextual bound of the full inequality satisfies $L/(L - 2d) \geq (D + 2)/(2\sqrt{D+1}) \sim 2^{N/2-1}$. Cabello [C] proposed this inequality in 2010 and argued for exponential growth; his Lemma 3 identifies the bound with the number of positive contexts, which is correct for two qubits but not for three, where 252 of the 315 predictions can be satisfied rather than 225 [M1]. Corollary A proves the growth.

Noise. If imperfections scale every correlator by a visibility η, the inequality is violated when $\eta > 1 - 2d/L$, hence whenever $\eta > 2\sqrt{D+1}/(D+2)$: above 0.6 for three qubits, 0.458 for four, 0.338 for five, 0.062 for ten. In the notation of [C, MS] the tolerated error per context $\varepsilon = 2d/L$ is at least $1 - 2\sqrt{D+1}/(D+2)$, which exceeds 0.707 (the largest value confirmed in the search of [MS], for configurations of a different kind) from six qubits on and tends to 1.

Proposition 4 (the floor is sharp up to a logarithmic factor). Let $N \geq 2$. For every pure N-qubit state with $M = \max_{v \neq 0} |\langle P_v \rangle|$ there is an assignment of the full space with

$$\beta \;\geq\; \frac{(D-1)(D-2)}{6\,M^3}, \qquad \text{so} \qquad \frac{1}{(D+1)(D+2)\,M_N^3} \;\leq\; \frac{1}{2} - \frac{d}{L} \;\leq\; \frac{\sqrt{D+1}}{D+2},$$

where M_N is the least value of M over all states. $M_N \geq 1/\sqrt{D+1}$, with equality exactly when a Pauli-covariant SIC exists, and $M_N \leq C\sqrt{\log D / D}$ for an absolute constant C.

Proof. A pure state has $\operatorname{tr}\rho^3 = 1$; with $\rho = D^{-1}(I + \sum_v p_v P_v)$ and $\sum_{v \neq 0} p_v^2 = D - 1$, the trace computation of Theorem 1 gives $T(p) = \sum_\ell s_\ell\, p_u p_v p_w = (D-1)(D-2)/6$ (the cubic purity condition [JL, St]). Choose the signs $f(v)$ independently with mean p_v/M. Then $\mathbb{E}\,\beta(f) = T(p)/M^3$, and some f does at least as well. The upper bound on M_N follows from Lévy's lemma for Haar-random states and a union bound over the $D^2 - 1$ observables [HLW]. ∎

So $\frac{1}{2} - d/L = \Theta(2^{-N/2})$ up to a factor $\log^{3/2} D$. In four sample runs the sign patterns of Haar-random states reached between 50 and 64 per cent of the floor value of β for four and five qubits. We did not find Proposition 4 in the notes we could read; we do not claim it is new.

Averaging between families. Every context of $W(7,2)$ lies on exactly 24 of its 120 elliptic quadrics, and every context of $Q^+(9,2)$ lies in exactly 112 of its 496 sections by an imaginary Pauli observable, each a signed copy of $W(7,2)$ [LGG, J1]; we re-verified these incidences by enumeration. With Theorem 2 and integer rounding this gives $d(Q^-(7,2)) \geq 291$, $d(W(7,2)) \geq 1455$, $d(Q^+(9,2)) \geq 6444$, $d(W(9,2)) \geq 28791$, each below the values of [J1, J4] and stated here

only to show what the floor alone yields.

7. The data against the floor

Table 2 confronts every published or pre-printed value we know for the classical families with the floor. "Floor" is the integer bound of Theorem 1 (Theorem 2 for elliptic quadrics), computed in exact integer arithmetic. "Best assignment" is the smallest number of violated contexts known; all entries up to five qubits were reproduced by our own search (Section 11).

Table 2. Classical configurations compared with the spectral floor.

configuration	N	points	contexts	averaging bound $\lceil L/5 \rceil$ [M2, LGG]	floor	best assignment	exact value
$W(3,2)$	2	15	15	3	2	3	3 [dB]
$Q^+(3,2)$	2	9	6	–	1	1	1
$W(5,2)$	3	63	315	63	**63**	63	63 [M1]
$Q^+(5,2)$	3	35	105	21	14	21	21 [dB]
$Q^-(5,2)$	3	27	45	9	**9**	9	9 [dB]
$W(7,2)$	4	255	5355	1071	1451	1575 [M2]	1575 [J1]*
$Q^+(7,2)$	4	135	1575	315	**315**	315 [M2]	315 [LGG]
$Q^-(7,2)$	4	119	1071	215	291	315 [M2]	315 [J1]*
$W(9,2)$	5	1023	86955	17391	28786	31479 [M2]	open ($\geq$ 28830 [J4]*)
$Q^+(9,2)$	5	527	23715	4743	6426	6975 [M2]	6975 [J1]*
$Q^-(9,2)$	5	495	19635	3927	6501	7087 [M2]	open ($\geq$ 6510 [J4]*)
$W(11,2)$	6	4095	1396395	279279	527621	553140 [M2]	open
$Q^+(11,2)$	6	2079	365211	73043	120901	132391 [M2]	open
$Q^-(11,2)$	6	2015	332475	66495	125624	131700 [M2]	open
$W(13,2)$	7	16383	22362795	4472559	9227609	9406024 [M2]	open
$Q^+(13,2)$	7	8255	5720715	1144143	2161542	2331191 [M2]	open
$Q^-(13,2)$	7	8127	5458635	1091727	2252409	2294580 [M2]	open

* computer-assisted, not formally verified, not peer reviewed; we did not re-run these computations.

Reading of the table. (1) All 17 rows are consistent with the floor; it is attained in the three rows in bold and, after rounding, in $Q^+(3,2)$, the Mermin–Peres square. (2) From four qubits on (five for hyperbolic quadrics) the floor exceeds the averaging bound $L/5$, and the gap to the best assignment shrinks: the floor closes 75, 81, 91 and 96 per cent of the interval between $L/5$ and the best assignment for the full space at $N = 4, 5, 6, 7$. (3) For seven qubits the best known assignments of the full space and of the elliptic quadric are within two per cent of the floor. (4) Outside these families the floor is exact for the configurations of Table 1.

An empirical regularity, also recorded for four qubits in [J1]: in every best assignment we computed, $A + 3I$ has low rank. The spectra of A are $\{21 \pm 16\sqrt{6}, (-3)^{14}\}$ for $W(7,2)$, $\{(9 \pm 4\sqrt{23})^2, (-3)^{12}\}$ for $Q^-(7,2)$, $\{45 \pm 56\sqrt{2}, (-3)^{30}\}$ for $Q^+(9,2)$, and three (respectively three pairs of) eigenvalues different from -3 for $W(9,2)$ (respectively $Q^-(9,2)$).

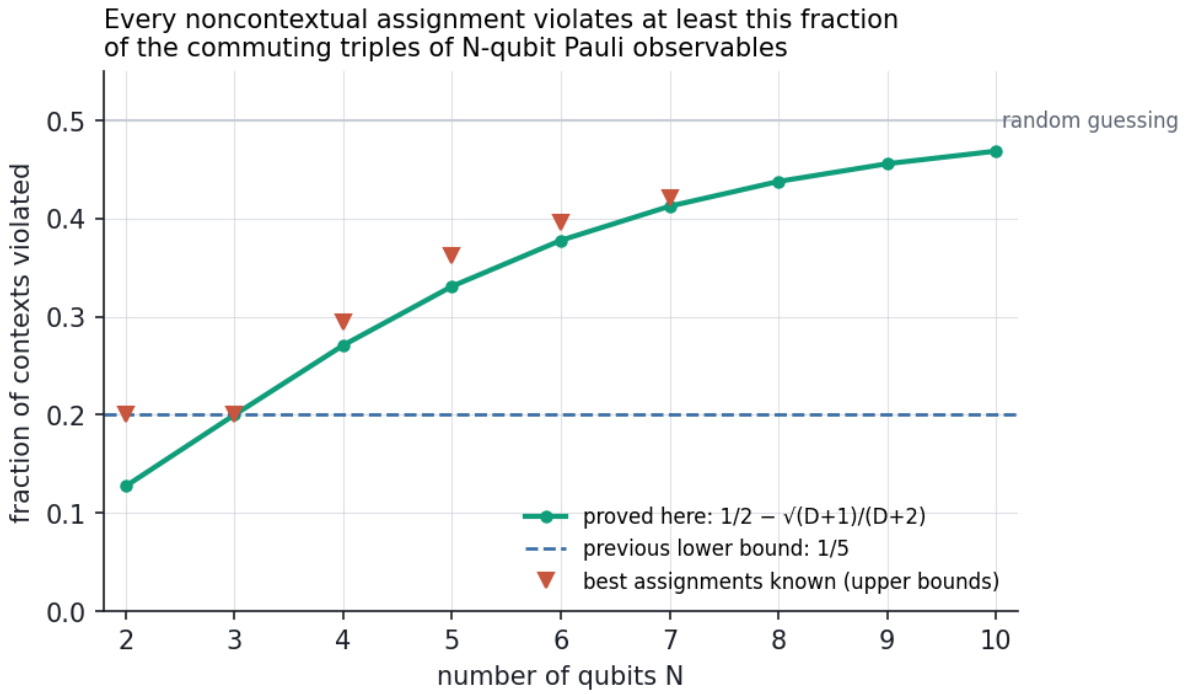


Figure 1. Lower bound on the fraction of violated contexts for the full N-qubit Pauli configuration, the earlier bound 1/5, and the best assignments known.

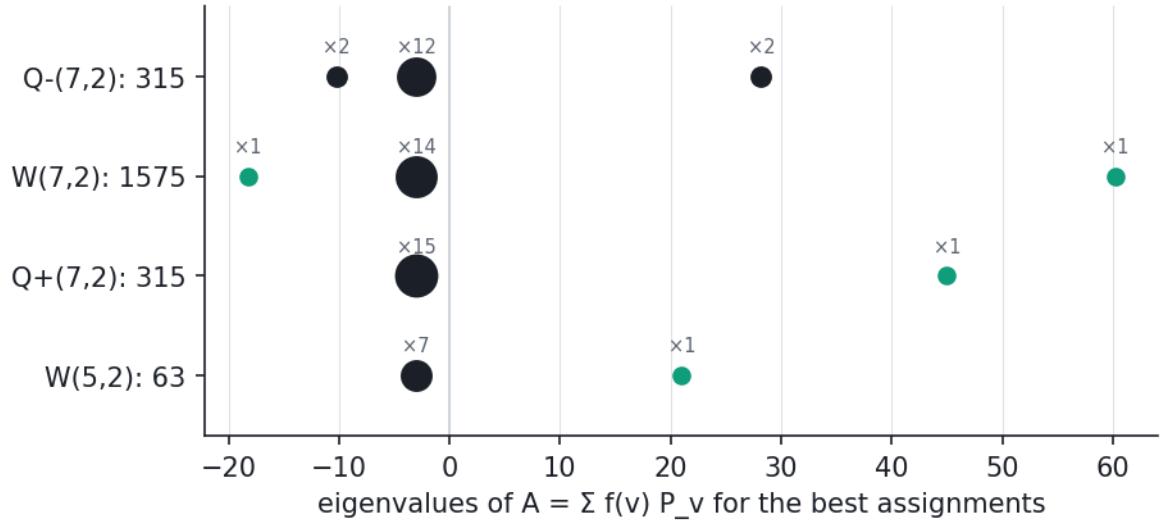


Figure 2. Spectra of $A = \sum f(v)P_v$ for the best assignments at three and four qubits.

8. What a carrier spectrum does not determine

8.1 The completion question

For a fixed assignment on a hyperplane section, the remaining contextuality constraints define a quadratic problem. Jonah's *strict law* says that equal spectra of the section carriers imply equal spectra of the corresponding completion operators within a specified class [J3]. Sign carriers, whose coefficients all belong to $\{-1, +1\}$, impose a stronger admissibility requirement than general real carriers: a construction using zero coefficients need not answer the sign-carrier question.

Case C3 in [J3] is a hyperbolic quadric cut by a hyperplane whose polar point lies on the quadric. Conjecture E predicts failure of the strict law on sign carriers in every vertex case for $N \geq 4$. The inspected source leaves this C3 range open. We give two admissible carriers at $N = 4$ and prove the predicted failure. This concerns the information in the entire carrier spectrum, even when its eigenvalues are known exactly; it leaves the spectral floor of Section 3 valid.

8.2 Geometry and two complete witnesses

Set $N = 4$, $D = 16$, and let $\mathcal{Q} = Q^+(7, 2)$ be the 135 nonidentity real Pauli observables, equivalently the labels with $q_0(a, c) = a \cdot c = 0$ modulo two. Use the polar point $p_\star = ZIII$ and write

$$H = \{v \in \mathcal{Q} : \langle v, p_\star \rangle = 0\}, \qquad O = \mathcal{Q} \setminus H.$$

Here H is the section and O its complement; the letter S retains its general meaning from Section 2. If R is the set of 35 real nonidentity three-qubit Paulis, then

$$H = \{Z \otimes I_8\} \cup \{I \otimes P, Z \otimes P : P \in R\}.$$

Thus $|H| = 71$ and $|O| = 64$. The outside points have first factor X with a real three-qubit factor, or first factor Y with an imaginary three-qubit factor: 36 and 28 points respectively. Each line not contained in H contains exactly one section point and two outside points. The quadric has 1575 contexts, of which 1120 are outside lines; these numbers refer to $\mathcal{Q}$, not S_B.

Define $\sigma_0, \sigma_1 : H \to \{-1, +1\}$ to be positive except at these labels:

Table 3. Complete C3 sign witnesses.

Carrier	Negative Pauli words	Negative numeric labels
σ_0	IIZZ, IYYI, ZIZZ, ZYYI	48, 102, 176, 230
σ_1	IYXY, ZIIZ, ZIZZ, ZYXY	87, 144, 176, 215

Both give $+1$ to $p_\star$. No search seed or external assignment file is needed to specify them. Their carrier operators are $A_\sigma = \sum_{s \in H} \sigma(s) P_s$. Define the real symmetric completion matrix Λ_σ, indexed by O, by

$$(\Lambda_\sigma)_{uv} = (\Lambda_\sigma)_{vu} = s_\ell \sigma(s) \quad \text{when } \ell = \{s, u, v\},\ s \in H,\ u, v \in O,$$

and zero for all other entries, including the diagonal. For $y \in \{-1, +1\}^O$ the number of violated outside lines is

$$F(\sigma, y) = 560 - \frac{1}{4} y^{\mathsf{T}} \Lambda_\sigma y. \tag{C1}$$

Each line contributes $(1 - s_\ell \sigma(s) y_u y_v)/2$; the symmetric quadratic form counts it twice. This proves the normalization and agrees with [J3, Section 2].

8.3 Exact statement

Theorem 14 (four-qubit C3 certificate). The witnesses in Table 3 have equal characteristic polynomials,

$$\chi_{A_{\sigma_0}}(t) = \chi_{A_{\sigma_1}}(t) = (t-3)^2(t+1)^8(t+5)^2(t+13) \cdot (t^3 - 25t^2 - 133t + 1621), \tag{C2}$$

but their completion operators satisfy

$$\operatorname{tr} \Lambda^3_{\sigma_0} = 14976, \qquad \operatorname{tr} \Lambda^3_{\sigma_1} = 15360. \tag{C3}$$

Consequently the carrier spectra, including multiplicities, are equal and the completion spectra are different. This proves failure of the strict law for four-qubit C3 sign carriers, establishing the corresponding instance of Conjecture E.

8.4 Block proof and the information that is lost

Every section operator commutes with $p_\star$, so in its ± 1 eigenspaces

$$A_\sigma = \operatorname{diag}(B_{\sigma,+}, B_{\sigma,-}).$$

Put $\alpha = \sigma(Z \otimes I)$, $a_P = \sigma(I \otimes P)$ and $b_P = \sigma(Z \otimes P)$. The two real symmetric 8×8 blocks are

$$B_{\sigma,+} = \alpha I + \sum_{P \in R} (a_P + b_P)P, \qquad B_{\sigma,-} = -\alpha I + \sum_{P \in R} (a_P - b_P)P. \tag{C4}$$

Since $a_P, b_P \in \{-1, +1\}$, each P occurs in exactly one block with coefficient $+2$ or -2. For three-qubit codes use $a + 8c$ and let $T_R = \sum_{P \in R} P$. Table 3 gives

$$\begin{aligned} B_{0,+} &= I + 2(T_R - 2P_{24} - 2P_{54}), & B_{0,-} &= -I, \\ B_{1,+} &= I + 2(T_R - P_8 - P_{24} - 2P_{47}), & B_{1,-} &= -I + 2P_8 + 2P_{24}. \end{aligned} \tag{C5}$$

Here $P_8 = IIZ$, $P_{24} = IZZ$, $P_{47} = YXY$ and $P_{54} = YYI$. Define

$$g(t) = t^3 - 25t^2 - 133t + 1621, \qquad h(t) = (t+13)g(t).$$

Exact determinant evaluation in (C5) yields the following finite integer identities.

Table 4. The block characteristic polynomials.

Block	Characteristic polynomial
$B_{0,+}$	$(t-3)^2(t+5)^2 h(t)$
$B_{0,-}$	$(t+1)^8$
$B_{1,+}$	$(t+1)^4 h(t)$
$B_{1,-}$	$(t-3)^2(t+1)^4(t+5)^2$

They can be reproduced directly from the explicitly specified eight-dimensional matrices; no approximate root comparison or classification theorem is involved. The last row also follows from the joint eigenvalues of its two commuting Z-type Paulis. Multiplying the plus and minus polynomials in each pair proves (C2).

The unlabelled union of block spectra stays fixed while their allocation changes. Two copies each of 3 and -5 move from the plus block to the minus block, and four copies of -1 move in the reverse direction. The common carrier spectrum alone does not record this allocation.

The outside Pauli span is the space of real symmetric off-diagonal block matrices

$$\begin{pmatrix} 0 & M \\ M^{\mathsf{T}} & 0 \end{pmatrix}, \qquad M \in \mathbb{R}^{8\times 8}.$$

One half of the anticommutator with A_σ acts there as

$$M \longmapsto \frac{1}{2}(B_{\sigma,+}M + MB_{\sigma,-}). \tag{C6}$$

Its matrix in the outside Pauli basis is exactly Λ_σ: commuting Pauli pairs supply the context sign and anticommuting pairs vanish. The polar term contributes zero on this outside span. This is the C3 Jordan representation of [J3, Theorem A and Table 1]. In orthonormal block eigenbases,

$$\operatorname{spec}\Lambda_\sigma = \left\{ \frac{b_{+,i} + b_{-,j}}{2} : 1 \le i, j \le 8 \right\}, \tag{C7}$$

where each ordered pair contributes once. These sums depend on the block allocation that the carrier spectrum has discarded. Expanding their cubes gives

$$\begin{aligned} \operatorname{tr}\Lambda_\sigma^3 = {} & \operatorname{tr} A_\sigma^3 \\ & + \frac{3}{8}\big[(\operatorname{tr} B_{\sigma,+}^2)(\operatorname{tr} B_{\sigma,-}) + (\operatorname{tr} B_{\sigma,+})(\operatorname{tr} B_{\sigma,-}^2)\big]. \end{aligned} \tag{C8}$$

For both witnesses $\operatorname{tr} B_+ = 8$, $\operatorname{tr} B_- = -8$ and $\operatorname{tr} A^3 = 18336$. If r real nonidentity three-qubit Paulis occur in the plus block, Pauli orthogonality gives

$$\operatorname{tr} B_+^2 = 8(1 + 4r), \qquad \operatorname{tr} B_-^2 = 8(1 + 4(35 - r)). \tag{C9}$$

For σ_0, $r = 35$, so these traces are 1128 and 8; for σ_1, $r = 33$, so they are 1064 and 72. Equation (C8) therefore gives

$$18336 + 3(8 - 1128) = 14976, \qquad 18336 + 3(72 - 1064) = 15360.$$

Equal spectra would have equal cubic traces. Their difference of 384 proves (C3) and the spectral conclusion without numerical diagonalization. ■

8.5 Consequence for the continuous completion bound

Corollary 15. The largest completion eigenvalues differ by exactly two:

$$\lambda_{\max}(\Lambda_{\sigma_1}) - \lambda_{\max}(\Lambda_{\sigma_0}) = 2. \tag{C10}$$

Let ρ be the largest root of g. The values $g(25) = -1704$, $g(30) = 2131$ and $g'(t) = 3t^2 - 50t - 133 > 0$ for $t \ge 25$ show that exactly one root lies above 25, and it belongs to $(25, 30)$. All roots are real because they occur in a real symmetric block polynomial. Both plus blocks have largest eigenvalue ρ; their other explicit eigenvalues are at most 3. The minus-block maxima are -1 and 3 respectively. Equation (C7) gives $(\rho - 1)/2$ and $(\rho + 3)/2$, proving (C10).

On the continuous sphere $y^{\mathsf{T}}y = 64$, equation (C1) has minimum

$$\min_{\|y\|^2=64} F(\sigma, y) = 560 - 16\lambda_{\max}(\Lambda_\sigma). \tag{C11}$$

This relaxed minimum is 32 lower for σ_1 than for σ_0. The binary minimum is bounded below by the spherical one; a binary vector need not be a top eigenvector. The corollary therefore does not prove a difference of 32, or any difference, between exact binary completion minima.

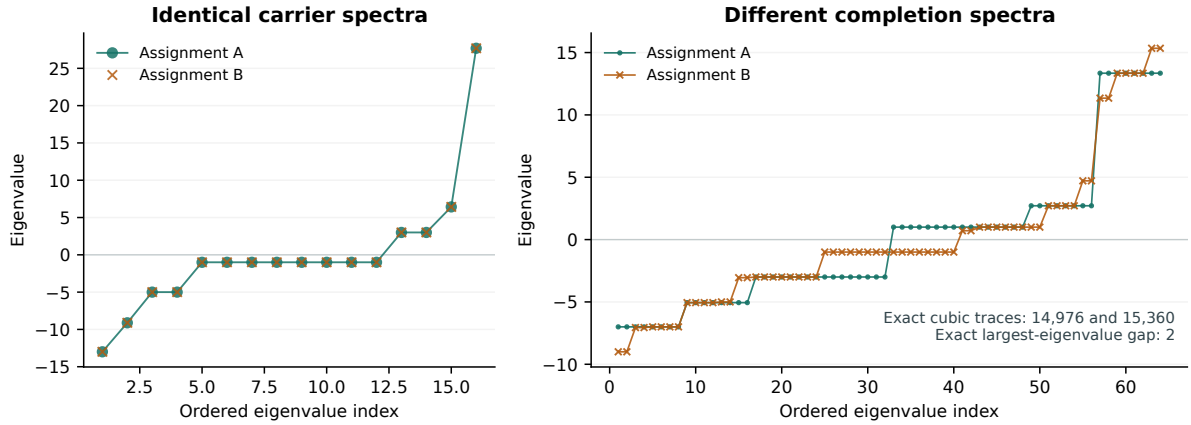


Figure 3. C3 carrier and completion spectra. Assignment A is σ_0 and B is σ_1. Numerical eigenvalue locations illustrate the certificate; exact integer identities establish carrier cospectrality, unequal completion cubic traces and the largest-eigenvalue gap. Multiplicities are retained.

The fixed-section score is also identical. The section has $1575 - 1120 = 455$ contexts. The trace identity of Section 3 gives $\beta_H = 18336/(6 \cdot 16) = 191$ for each carrier, hence $(455 - 191)/2 = 132$ violated section contexts and 323 satisfied ones. Independent integer Pauli-label phase arithmetic in `scripts/unified_section_score.py` recounts these values; the record is `results/unified-section-score.json`. Flipping label 1 of the first assignment changes its count to 147 and its score to 161, furnishing a negative control. This section-score observation is a derived and executable check, not an additional specialized Lean theorem.

The floor uses spectral moments of an assignment on a whole chosen configuration to certify a universal inequality. The strict law asks whether a section carrier's eigenvalues encode a different operator. The block partition explains why the first method remains valid when the second inference fails. Neither the known degree 315 of $\mathcal{Q}$ nor the degree 75 of S_B is altered by this completion example.

9. Formal verification and its boundary

The sources are in `lean/HoggarBound/` (Lean 4.35.0, Mathlib v4.35.0).

9.1 Floor and saturation statements

module	content
`Moment.lean`	`sum_cube_le` (the scalar inequality); `trace_pow_eq_sum_eigenvalues_pow` ($\operatorname{tr} A^k = \sum \lambda_i^k$ for Hermitian A); `trace_cube_le`
`Pauli.lean`	bit vectors `Fin N → ZMod 2`; Weyl operators and `pauli u` as explicit complex matrices; `pauli_mul`, `context_product` ($P_u P_v P_{u+v} = \tau(u,v) I$), `pauli_comm`; traces of one, two and three Pauli operators; $\tau = \pm 1$ on commuting pairs

module	content
`Bound.lean`	`flags` (ordered commuting pairs with product in S: six per context), `beta6`, the operator `opA`; `trace_opA_cube` ($\operatorname{tr} A^3 = D\,\beta_6$); `hoggar_bound`
`Counting.lean`	`violated`, `beta6_eq_card`; the flag count of the full space for every N (`card_flags_all`); `violatedLines` and `card_violated_le`; `four_qubit_contexts`
`Corollaries.lean`	`full_space_bound`, `full_space_fraction` (all N); `three_qubit_contexts`, `five_qubit_contexts`
`Quadric.lean`	the real quadric; its point and flag counts for every N by character sums; `four_qubit_hyperbolic_contexts`, `five_qubit_hyperbolic_contexts`
`MaskConfig.lean`	configurations and assignments given by bit masks; `card_flags_cfg`, `card_violated_cfg` (their flag counts as sums the kernel can evaluate); `degree_of_certificate`
`FlatData.lean`, `SaturatedFive.lean`	generated: the masks of the first six rows of Table 1, and the five-qubit counts in 32 blocks
`Saturated.lean`	`pivot_degree`, `cieslinski5_degree`, `quadric_degree`, `hoggar_degree`, `ccz_degree`, `lly_degree`
`PhaseData.lean`, `SaturatedPhaseFive.lean`	generated: the masks of Φ_5 and Φ_7; `phase5_degree`

The main theorem, as checked:

```
theorem hoggar_bound (hN : 1 ≤ N) (S : Finset (Pt N)) (h0 : (0 : Pt N) ∉ S) (f : Pt N → ℝ)
    (hf : ∀ u ∈ S, f u = 1 ∨ f u = -1) :
    ((2 : ℝ) ^ N - 1) * beta6 S f
      ≤ ((2 : ℝ) ^ N - 2) * S.card * Real.sqrt (((2 : ℝ) ^ N - 1) * S.card)
```

Here `beta6 S f` is six times $\beta_S(f)$, a sum over ordered pairs, and the sign of a context is by definition the real part of the scalar $\tau(u, v)$ with `pauli u * pauli v * pauli (u + v) = tau u v • 1`. Table 5 lists the corollaries.

Table 5. Formal floor and exact-degree statements.

Lean theorem	statement
`full_space_fraction`	for all $N \geq 1$: #violated flags $\geq \left(\frac{1}{2} - \sqrt{2^N+1}/(2^N+2)\right)$ #flags
`three_qubit_contexts`	every assignment of the 63 observables violates at least 63 contexts
`four_qubit_contexts`	every assignment of the 255 observables violates at least 1451 contexts
`five_qubit_contexts`	every assignment of the 1023 observables violates at least 28786 contexts
`four_qubit_hyperbolic_contexts`	every assignment of the 135 real observables violates at least 315 contexts
`five_qubit_hyperbolic_contexts`	every assignment of the 527 real observables violates at least 6426 contexts
`pivot_degree`	for the 135 observables of S_B: every assignment violates at least 75 contexts, and some assignment violates exactly 75
`cieslinski5_degree`	for the 496 observables of S_5: at least 480, and exactly 480 for some assignment
`phase5_degree`	the same statement for the 496 observables in the support of the circuit state Φ_5

Lean theorem	statement
`quadric_degree`, `hoggar_degree`	the same two-sided statements with 315 (four-qubit hyperbolic quadric) and 63 (three qubits)
`ccz_degree`, `lly_degree`	some assignment of the 28 (respectively 12) observables violates no context

`#print axioms` reports `[propext, Classical.choice, Quot.sound]` for each of them: no `sorry`, no `native_decide`, no additional axiom.

How the exact degrees are checked. The statement of `pivot_degree` is about a set of 135 explicit complex 16×16 matrices, given by a 256-bit mask: bit $a + 16c$ says whether $P_{(a|c)}$ belongs to it. `MaskConfig.lean` proves, for every N and every mask, that the number of flags of the configuration and the number of flags violated by an assignment that is itself given by a mask are double sums over codes of Boolean expressions built from bit tests, exclusive or and addition of natural numbers. The kernel evaluates these sums (`decide +kernel`); for five qubits each sum has 2^{20} terms and is evaluated in 16 blocks. `degree_of_certificate` then combines three evaluated numbers (observables, flags, flags violated by the witness) with `hoggar_bound`. That the masks are the supports and sign patterns of the states of Table 1 is checked by the programs of Section 11 in exact arithmetic; it is not needed for, and not part of, the Lean statements. For seven qubits the two sums have 2^{28} terms each and were not evaluated in the kernel: the degree 729 344 of S_7 rests on `hoggar_bound`, which needs only the number 8128, and on a count of contexts and violated contexts made by programs.

Floor and saturation results outside the formalization. Theorem 2 (the quaternionic form) and all elliptic-quadric numbers; Theorems 3 and 4 as equivalences, and the state, spectral-equality and geometric interpretations in Sections 4–6 (the Lean results for that part are the inequality and the two-sided degree statements); Propositions 4, 6 and 8 to 11; the enumerations of Theorems 12 and 13 and the seven-qubit counts; the limit statement as a theorem about limits (the inequality for each N is formal); the exact values of [J1]. For the classical families beyond three qubits and $Q^+(7, 2)$, upper bounds are witnesses checked by the programs of Section 11.

Independent audit. A separate set of agent sessions, working concurrently in the same repository and without editing our files, re-checked the first six modules from source, confirmed the axiom report, and added two modules of its own (`IndependentContexts.lean`, `IndependentWitness.lean`). They prove that a violated context carries exactly six violated flags (used in `MaskConfig.lean`) and, by kernel evaluation of an integer certificate, that the explicit assignment "$f(a, c) = +1$ if $a = 0$ or $c = 0$ or the number of Y factors is at least 2, and -1 otherwise" violates exactly 63 contexts. Their reports are in `results/lean-audit.md`, `results/literature-audit.md` and `results/stack-independent-audit.md`, and their companion note is `paper/independent-research-note.md`. The literature audit is what first pointed us to [J1]. The modules from `MaskConfig.lean` on, and everything in Section 5, were written after that audit and are not covered by it.

9.2 The checked C3 statement

The C3 development uses the same pinned Lean 4.35.0 and Mathlib 4.35.0 setting. It identifies the complete 71-point section and 64-point complement, proves the assignments are sign-valued, and connects the integer certificate matrices to the original complex Pauli sums and real `lineSign` completion operators. Thus its conclusion applies to the configuration of Section 8.

The main declarations in `HoggarBound.C3` are `four_qubit_C3_counterexample`, `pauli_carrier_eigenvalues_eq`, `pauli_completion_eigenvalue_multisets_ne` and `four_qubit_C3_spectral_c`

ounterexample. They prove equal carrier characteristic polynomials and canonical Hermitian eigenvalue lists, different completion characteristic polynomials and real root multisets with multiplicities, and the exact completion cube traces 14976 and 15360.

The certificate kernel-checks $A_0M = MA_1$ and $MR = 46656I$ for explicit integer matrices. This certifies an invertible similarity over the rationals. For each completion, an intermediate square is checked entry by entry and substituted into a checked cubic trace. No external eigensolver supplies a proof premise. All 42 declarations in `lean/C3Verification.lean` passed the axiom audit with only `propext`, `Classical.choice` and `Quot.sound`; the finite checks use kernel reduction, not `native_decide`. Commands, source hashes and results are in `results/frontier-c3-lean-audit.md` and its evidence files.

Table 6. Boundaries that matter for interpreting the combined result.

Claim	Evidence boundary
Arbitrary-support spectral floor and stated finite corollaries	Lean statements about the defined Pauli/context model
Degrees 75 and 480 for the explicit masks	Lean lower bounds, witnesses and kernel-evaluated counts; state-to-mask interpretation checked separately by programs
Saturation equivalences, phase-state classification and seven-qubit counts	Mathematical arguments and executable computations; not general Lean equality/classification theorems
Four-qubit C3 carrier cospectrality, unequal completion spectra and cube traces	Complete 42-declaration Lean audit, including original Pauli and context-sign bridges
Printed C3 block/carrier factorizations, largest-eigenvalue gap 2 and spherical gap 32	Exact algebraic and executable checks; not included in those 42 Lean declarations
Physical hardware, compiler correctness, historical priority or external peer review	Not established by these theorem checks

The floor, saturation and C3 audits have different scopes. The independent audit of the original six floor modules does not retroactively cover the later saturation development, and the C3 audit does not certify the phase-state enumeration. Combining the manuscripts introduces no new Lean theorem and does not broaden any existing theorem's hypotheses or conclusion.

Successful CI provides a second recorded execution at commit `87f08a0`. The full verification run passed the executable checks and audits of 42 C3, 22 baseline and six auxiliary stability declarations. The separate saturation run passed its exact-computation jobs and 14-declaration audit. The archived records `results/full-ci-87-audit.json` and `results/saturation-ci-audit.json` identify the checkout, artifact digests and source hashes. These are checks of those source snapshots, not evidence of a new proof produced by combining the texts; the auxiliary stability results are not used here.

10. A running example on the Tsotchke stack

10.1 State preparation, simulated contexts and quantum geometry

All four components are used for what they are.

- **Moonlab** (v1.2.1, prebuilt WebAssembly engine) is a classical state-vector simulator. It plays the quantum device.
- **quantum_rng** is a classical, "quantum-inspired" integer and trigonometric mixer, not a hardware quantum source. It supplies seeded pseudo-random numbers. Its stream

depends on the C library it is built against, so no result depends on its exact values: every count is recomputed independently.

- **QGTL** (quantum_geometric_tensor): only the `diffgeo_*` kernels (Fubini–Study metric, Berry curvature), which we had verified earlier against closed forms, are used.
- **Eshkol** (v1.3.6) is a Scheme dialect with automatic differentiation. It carries an independent implementation of the geometry and the differentiation checks.

Experiment on Moonlab, classical families (`src/moonlab_experiment.mjs`, seed 20261109). A Pauli observable is measured projectively by rotating its eigenbasis onto Z of one qubit (gates H, $S^\dagger$, CNOT), calling Moonlab's single-qubit measurement with a number from quantum_rng, and undoing the rotation.

step	what is done	result
E0	Moonlab's $\langle P \rangle$ against a direct evaluation; repeated measurement	agree to 8.6×10^{-16}; repeats agree
E1	ψ_H on 3 simulated qubits; 63 expectation values; signs as an assignment	all $\pm 1/3$ (to 6×10^{-16}); 63 of 315 contexts violated, three through each point
E2	realified ψ_H on 4 simulated qubits	$\pm 1/3$ on 135 real observables, 0 on 120 imaginary ones; 315 of 1575 violated, seven through each point
E3	every context measured sequentially on random states, $N = 2, 3, 4$, three runs each	15/15, 315/315, 5355/5355 products equal to the quantum sign (16065 projective measurements and 161406 gates per four-qubit run)
E4	the same with a random Pauli error of probability p on each qubit before each measurement; 3000 to 4000 sampled contexts	Figure 4 and the table below

The table reports the fraction of sampled contexts whose outcomes multiply to the quantum sign, against proved noncontextual ceilings. For three and four qubits the ceiling is $1 - d_{\text{floor}}/L$. The two-qubit column uses the independently known exact degree $d = 3$, giving $1 - 3/15 = 0.800$; the spectral floor alone rounds to 2 there.

error p	$N = 2$	$N = 3$	$N = 4$
0	1.000	1.000	1.000
0.01	0.979	0.969	0.957
0.02	0.958	0.939	0.930
0.05	0.897	0.867	0.830
0.10	0.805	0.747	0.717
0.20	0.669	0.616	0.591
ceiling, proved	0.800	0.800	0.729
best classical model known	0.800	0.800	0.706

Experiment on Moonlab, the configurations of Table 1 (`src/moonlab_saturated.mjs`, seed 20261110).

step	what is done	result
S1	the first six states of Table 1 loaded as amplitudes; Φ_5 and Φ_7 prepared by running their circuits (Hadamard, controlled-phase and Toffoli gates of Moonlab: 14 and 34 gates); every Pauli expectation value read through Moonlab's gates	the circuits give the phase states to 10^{-16}; non-zero expectation values of one modulus to 10^{-15}; supports and signs equal the masks in `data/flat/`; contexts and violated contexts as in Table 1, including 2 824 192 and 729 344 for seven qubits
S2	every context of S_B (three runs), of S_5 and of the support of Φ_5 measured sequentially on random states; for S_7 a sample of 30000 contexts, and once all of them	1095/1095, 10880/10880, 10880/10880, 30000/30000 products equal to the quantum sign; in the full seven-qubit run 2 824 192/2 824 192 (8 472 576 projective measurements, 161 550 336 gates)
S3	the same with Pauli noise; 4000 sampled contexts	the table below

Because the degrees are exact, the ceilings in this table are exact as well: $1 - d/L$. For the four- and five-qubit configurations they are much higher than for the full space, so the test tolerates much less noise; for S_7 the ceiling is 0.742 and the simulated test stays above it up to an error probability of two per cent per qubit and measurement.

error p	S_B (four qubits)	S_5 (five qubits)	support of Φ_5	S_7 (seven qubits)
0	1.000	1.000	1.000	1.000
0.002	0.992	0.989	0.990	0.986
0.005	0.982	0.981	0.978	0.963
0.01	0.962	0.955	0.949	0.938
0.02	0.929	0.913	0.916	0.877
0.05	0.835	0.807	0.797	0.733
0.10	0.719	0.679	0.670	0.597
ceiling, exact	0.932	0.956	0.956	0.742

These are frequencies in a seeded software simulation. They illustrate the inequalities and are not measurements on hardware or a loophole analysis.

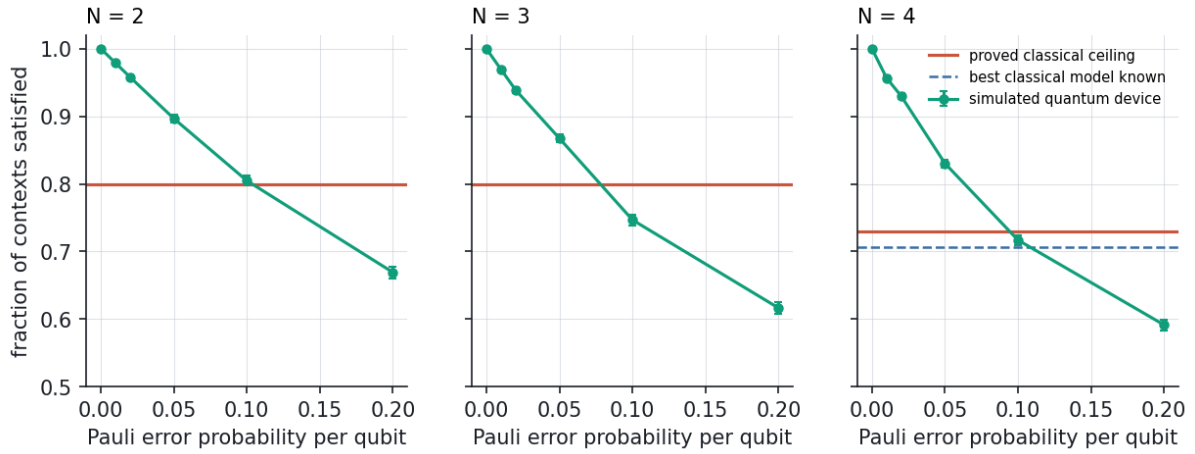


Figure 4. Simulated success fraction on Moonlab against the classical ceilings, for the full space.

QGTL (`src/qgt_hoggar.c`, `src/qgt_flat.c`). The kernels `diffgeo_compute_fubini_study` and `diffgeo_compute_berry_curvature`, applied to ψ_H and the 63 tangent vectors $-\frac{i}{2}P_v\psi_H$, return the values of Proposition 3 to 10^{-16}, and the 189 pairs with $|g_{uv}| > 1/12$ are exactly the pairs

on the 63 violated contexts. For the eight states of Table 1 and the tangent vectors of their supports they return the three values of Proposition 8 to 10^{-16}; the pairs above the threshold recover the 75 violated contexts of S_B (225 pairs of 9045), the 480 of S_5 and of the support of Φ_5 (1440 pairs of 122760), and the 729 344 of S_7 (2 188 032 pairs of 33 028 128), and no others.

Eshkol (`src/eshkol/hoggar.esk`, `src/eshkol/saturated.esk`). An exhaustive pass over all 2^{15} two-qubit assignments gives minimum 3, attained 640 times. The Hoggar signs violate 63 contexts. Reverse-mode automatic differentiation of the cubic form at the Hoggar Pauli vector x^* gives $T(x^*) = 7 = (D-1)(D-2)/6$ and $\nabla T(x^*) = 3x^*$ to 7×10^{-16} (the Lagrange condition for a maximizer on the sphere); nested forward-mode second derivatives along four tangent directions are negative (-0.59 to -0.74). For the pivot state the second script finds 135 non-zero expectation values, 1095 contexts and 75 violated ones, $T = 35 = (D-1)(D-2)/6$, and $\nabla T = 7x$ to 1.3×10^{-15}, which is condition (d) of Theorem 4.

10.2 Exact C3 arithmetic in Eshkol

The separate serial compiled run uses Eshkol 1.3.6-evolve through `eshkol-run -r`. It checks $A_0 M = M A_1$ and both $MR = RM = 46656I$, then evaluates the block-moment formula to obtain 14976 and 15360. An arbitrary-precision Python replay bounds every input, product and ordered accumulation of this program by 408304 in absolute value, below $2^{63} - 1$. The recorded run returned zero with empty standard error; `results/frontier-c3-eshkol.json` records its command, hashes and process-local paths to the existing LLVM 21 and CUDA 12.9 installations. This qualifies the tested serial arithmetic execution, not compiler correctness or GPU/parallel behavior.

11. Falsifiability and replication

11.1 Floor, saturation and simulated measurement checks

Predictions that could fail. (P1) No assignment violates fewer contexts than the floor. (P2) The saturated cases are reached exactly. (P3) For every S and f, $\operatorname{tr} A = 0$, $\operatorname{tr} A^2 = D|S|$, $\operatorname{tr} A^3 = 6D\beta$. (P4) On an elliptic quadric the spectrum of A is doubly degenerate. (P5) For every pure state $T(p) = (D-1)(D-2)/6$. (P6) Ideal sequential measurements reproduce every context sign in every state. (P7) For a flat state, condition (d) of Theorem 4 holds. A counterexample to P1 would provide a finite assignment whose violation count can be recomputed independently from the context definitions.

Tests (`scripts/run_all.sh`).

- `hoggar exhaust2`: all 32768 two-qubit assignments; minimum 3 (640 assignments); for the square, minimum 1.
- `scripts/run_falsify.sh`: 20 runs over $N = 2, \ldots, 5$ and the three families; 20000, 5000 and 500 uniformly random assignments and 1500, 200 and 12 annealed ones per configuration for $N = 3, 4, 5$. No assignment below the floor. Minima reached by annealing: 63, 21, 9 ($N = 3$); 1575, 315, 315 ($N = 4$); 31479, 6975, 7087 ($N = 5$). These equal the best known values in every case.
- A negative control: the false claim $d(W(5,2)) \geq 64$ is refuted by the Hoggar witness with exit code 3, showing that the harness can fail.
- `scripts/flat_states.py` (68 checks, integer arithmetic in $\mathbb{Z}[i]$ except one labelled check): for the six states of Table 1, the single modulus, the support, the number of contexts, the number violated, the floor, condition (d) at every point and the vanishing of every outside sum; the statements of Theorem 5 about S_B; Proposition 6 on all 4096 phase patterns;

Proposition 9 for one to four qubits; and an annealing search from random starts on S_B and S_5, which reaches 75 and 480 and never goes below.

- `build/third_level` (seven seconds): the enumeration of Theorem 12, all 33 554 431 forms for five qubits included, with the orbit computation; it fails if any count differs from the table.
- `scripts/phase_states.py` (81 checks, integer arithmetic): for $CS|{+}{+}\rangle$, $CCZ|{+}{+}{+}\rangle$, $CS\,CS|{+}{+}{+}\rangle$, Φ_5 and Φ_7, the constant rank, the single modulus, the support, the numbers of contexts and violated contexts, condition (d) and the outside sums, and the description of the support in Proposition 11; Proposition 10 for all eleven states of this paper; the enumeration of Theorem 12 for $N \leq 4$ by a second implementation; the two symplectic maps of Theorem 12; and the finite-field descriptions of the five- and seven-qubit forms.
- `scripts/run_family7.sh` (about eight core-hours, not part of the everyday run): the enumeration of Theorem 13, the orbit computation, and the random census of the other types; its output is `results/family7.log`. The two counts agree: 13 271 040 solutions found, and $92\,897\,280/7 = 13\,271\,040$ for one orbit with a stabilizer of order 7.
- The seven-qubit numbers 8128, 2 824 192 and 729 344 are computed three times without shared code: in Python with exact integers, in C from the masks (inside the QGTL program) and in JavaScript from the expectation values returned by Moonlab.
- `scripts/analysis.py` (62 checks) and `scripts/ladder.py` (14 checks): P3 on random data to machine precision; P4 on all 120 four-qubit elliptic quadrics; P5; the geometry of the violated sets; closed-form counts against enumeration; agreement of the C and Python integer bounds.
- Witness files (`data/witness/`, `data/flat/`) are recounted by independent routines that use no tables.

Independent implementations. The geometry, the context signs and the counts are implemented five times: C (`src/hoggar.c`), Python with explicit matrices or exact integers (`scripts/analysis.py`, `scripts/flat_states.py`), JavaScript (`src/moonlab_experiment.mjs`, `src/moonlab_saturated.mjs`), Eshkol, and the Lean model, in which the signs are theorems about matrix products. Their overlapping checks agree on the counts recorded for those tests. This does not mean that each implementation checks every result: in particular, the seven-qubit context counts are outside the kernel evaluation, and the C3 checks use the separate programs described next.

11.2 Independent C3 reconstruction and negative controls

The literal-matrix verifier builds each four-qubit Pauli from the 2×2 matrices, selects the real quadric and section, and computes exact carrier characteristic polynomials. A change-of-basis matrix T with $T^{\mathsf{T}}T = TT^{\mathsf{T}} = 8I$ intertwines the completion matrix and the block Jordan representation, checking that the coordinate transformation loses no component.

A separate verifier constructs the completion matrix from all 1120 outside signed contexts. It checks commutation and multiplies each triple to determine its sign, then recomputes both carrier polynomials and completion cube traces. This route does not use (C7). Its last three tensor factors are reversed relative to the printed convention; that fixed orthogonal basis permutation is recorded explicitly and preserves the represented assignments and spectra. Pauli entries $0, \pm 1, \pm i$ are exact in their initial representation, real matrices are converted to integers, and product bounds fit signed 64-bit arithmetic. Characteristic polynomials use exact SymPy arithmetic, not eigenvalue tolerances.

Flipping one off-diagonal context entry and its symmetric partner changes the two cubic traces to 14808 and 15144 respectively, so the expected certificate fails. The verifier rejects Python

`-O`, which would disable its assertions. These are specified negative controls, not a proof that every possible implementation error is detected.

The interactive C3 model independently constructs context signs, computes characteristic polynomials with a BigInt Faddeev–LeVerrier recurrence and evaluates cube traces with BigInt arithmetic. Editing a sign triggers a fresh computation. A modified assignment is not required to retain cospectrality; the original four-negative-word witnesses are the exact claim being tested. The experiments concern finite mathematical operators and are separate from the Moonlab measurement simulations.

12. Limitations and open problems

- Novelty is claimed to the extent of a bounded search (arXiv and Zenodo queries on 9 November 2026, and the sources of Section 1.2). The field moved quickly in October 2026, and several of the relevant preprints are less than a month old. Statements of the form "to our knowledge" mean exactly that.
- The configurations S_B, S_5 and S_7 are defined by states, not by a classical geometric construction. We do not know an infinite family of non-closed configurations that attain the floor: Proposition 9 shows that the simplest family of states stops at four qubits. In the nonzero-form odd-qubit phase-state family of constant rank $N-1$, the examples here have three, five and seven qubits; the rank-eight nine-qubit case is open. Theorem 12 is complete for its class of states; whether Table 1 is complete for all states on $N \leq 5$ qubits is supported by numerical search only. Whether the states of the odd class of Proposition 6 form a single Clifford orbit was not determined.
- For seven qubits the uniqueness statement is proved only inside the family $\mathcal{F}$ of Theorem 13; outside it there is a random census. The enumeration behind it (and the one of Theorem 12) is a computation, repeated here by one program only, apart from the agreement of the count with the orbit size and the second implementation for $N \leq 4$. The seven-qubit counts of contexts are checked by three programs but not by the Lean kernel.
- Within the classical families the floor is not attained beyond the three listed cases [J2]. Exact values for five or more qubits are open except $Q^+(9,2)$ [J1]; the floor alone leaves the intervals of Table 2.
- The external exact values $d(W(7,2)) = 1575$, $d(Q^-(7,2)) = 315$ and $d(Q^+(9,2)) = 6975$ rest on computer-assisted work not formally verified or peer reviewed in [J1]. This does not include $d(Q^+(7,2)) = 315$, which has the Lean proofs described above. A formal proof of $d(W(7,2)) = 1575$ would be a natural next target; our Lean development gives the lower bound 1451 only.
- The quaternionic form (Theorem 2) and the equivalences of Theorems 3 and 4 are proved on paper and tested numerically, not formalized.
- The C3 certificate settles only the four-qubit instance of the strict-law failure. It neither proves Conjecture E for all dimensions nor classifies all carriers. The exact binary completion minima of the two witnesses are not calculated here; the gap 32 belongs to a continuous relaxation. The completion result does not invalidate the spectral floor or revise the contextuality degrees.
- The literature comparison is bounded and qualitative. An exact finite certificate proves its stated mathematical conclusion, not that no earlier author knew it or that a venue will accept it.
- Nothing here is an experiment on quantum hardware, and nothing shows a computational advantage. The simulation demonstrates what quantum mechanics predicts and what no noncontextual model can do.

13. Research provenance, disclosure and availability

How the work was produced. Alexander S. Donahue initiated and directed the project and specified the Tsotchke stack. The floor, saturation, phase-state and associated simulation work was developed by Claude Fable 5 (Anthropic) through Claude Code. Within the author's brief that system selected problems, searched sources, derived results, wrote and ran programs and Lean proofs, and drafted the spectral-floor manuscript. Its development and verification records remain with that component.

OpenAI Codex research and coding agents performed the independent prior-art and original-floor audits, selected the finite C3 question, discovered its witnesses, developed the algebraic arguments and Lean formalization, implemented the independent checks and demonstration, and drafted the C3 manuscript and figures. Codex agents prepared this combined text, shared notation, section map and publication layouts. The combined presentation reuses the two established mathematical components; it does not create a new experimental dataset or claim a new unified Lean proof.

AI disclosure statement. Claude and Codex were used for substantive research, software, formalization and writing, not merely language correction. The recorded checks comprise Lean kernel verification, exact and numerical programs with their stated scopes, and source comparisons by AI agents. They do not constitute independent human peer review. No completed human critical review or venue-specific final approval is asserted here; the author must accurately describe that work before making a submission that requires it. AI systems are disclosed as tools, not listed as authors.

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Data availability statement. Source, data, formal proofs, test results and the earlier manuscripts are available in Hoggar Bound Lab. Release and workflow records identify the checked snapshots; earlier Software Heritage and Internet Archive captures remain historical records, not evidence that this combined revision has been archived or reviewed. Appendix A gives reproduction commands. No new arXiv identifier, DOI or journal decision is implied by this manuscript or its local submission packages.

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Appendix A. Reproduction

```
# engine, random source, QGT programs (gcc; no other dependency)
bash scripts/build.sh
# every test: exhaustive N=2, falsification suite, flat states, analysis, ladder, Moonlab, QGTL,
↪  Eshkol
```

```
bash scripts/run_all.sh
# Lean (needs elan; fetches Lean 4.35.0 and the Mathlib cache)
cd lean && lake exe cache get && lake build
# the exact degrees of Section 5 (kernel evaluations; see the CI evidence for time and memory)
lake build HoggarBound.Saturated HoggarBound.SaturatedPhaseFive
lake env lean SaturatedVerification.lean
# the seven-qubit enumeration of Theorem 13 (about eight core-hours)
cd .. && bash scripts/run_family7.sh
```

The C3 verification programs require Python, NumPy, SymPy and Node.js. From the repository root:

```
python scripts/frontier_c3_verify.py
python scripts/frontier_c3_context_verify.py
node tests/frontier_c3_browser_test.cjs
python scripts/frontier_c3_integer_bounds.py
python scripts/unified_section_score.py
python scripts/frontier_c3_demo.py
```

The last command requires Eshkol; set `ESHKOL_RUN` when it is not on the system path. The wrapper generates `build/frontier_c3_demo.esk` and invokes `eshkol-run -r`, retaining its process-local settings in the result record. The published similarity certificate, not a random search, is the input to these arithmetic checks.

From `lean/`, build the C3 targets sequentially and then audit the declarations:

```
lake build HoggarBound.C3Data
lake build HoggarBound.C3Certificate
lake build HoggarBound.C3Geometry
lake build HoggarBound.C3Theorem
lake build HoggarBound.C3Spectra
lake env lean C3Verification.lean
```

The recorded Windows finite-certificate check took approximately 61 minutes. This is proof-checking cost, not the running time of the interactive demonstration or a physical quantum process. The direct context check used Python 3.9.13, NumPy 2.0.2 and SymPy 1.14.0. Raw-byte and LF-normalized input hashes distinguish local Windows line endings from Git-normalized text.

Appendix B. Witness assignments

`data/witness/nN_t.txt` holds, for N qubits and family t (`f` full, `h` hyperbolic, `e` elliptic), one header line and a hexadecimal mask whose bit v is set when $f(v) = -1$, with $v = a + 2^N c$. The three-qubit Hoggar witness produced by the Moonlab run is

```
3 f 63
163e5efe76fefe00
```

`build/hoggar verify N t file` recounts any witness from the definitions.

`data/flat/nN_name.txt` holds the configurations of Table 1: a header line with N, the numbers of observables and contexts and the degree; the mask of the configuration; and the mask of the optimal assignment. For the pivot configuration:

```
4 135 1095 75
b4451eef7889d22344b5ee1f887922d34b45e1ef87892d23bbb5111f7779ddd2
20011cc01888c2204424221f001100c34244212f01010c038880111177790000
```

The same two numbers are `pivotMask` and `pivotNeg` in `lean/HoggarBound/FlatData.lean`. The files `n5_phase5.txt` and `n7_phase7.txt` hold the supports of Φ_5 and Φ_7 in the same format; the seven-qubit masks have 4096 hexadecimal digits.

For C3, the four negative labels of each carrier are given in Table 3. The principal machine-readable records are `results/frontier-c3-n4-pair.json`, `results/frontier-c3-n4-verification.json`, `results/frontier-c3-n4-completions.json` and `results/frontier-c3-context-verification.json`. They record the matrices, counts, exact polynomials and moments, mutation controls and dependency versions. The theorem's assignments are fully defined in Section 8.2 independently of those files.